Scientific Reports

https://doi.org/10.1038/s41598-025-30241-9

Article in Press

A complex circular intuitionistic fuzzy decision framework for evaluating sustainable agriculture strategies under uncertainty

Received: 8 August 2025

Accepted: 24 November 2025

Published online: 03 December 2025

Cite this article as: Yu Y. A complex circular intuitionistic fuzzy decision framework for evaluating sustainable agriculture strategies under uncertainty. *Sci Rep* (2025). https://doi.org/10.1038/s41598-025-30241-9

Yanfeng Yu

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

A Complex Circular Intuitionistic Fuzzy Decision Framework for Evaluating Sustainable Agriculture Strategies under Uncertainty

Yanfeng Yu[1,*]

[1]Institute of Agricultural Economics and Information, Jiangxi Academy of Agricultural Sciences, Nanchang, 330200, China

Correspondence: Yanfeng Yu (Email: fionayu6@163.com)

Abstract: Agricultural productivity is accompanied by complex multi-factor problems that are economic, environmental, and social. Decision-making practice in this area requires a methodology that can handle uncertainty, regular growth, circular patterns, and conflicting evaluation requirements. A new multi-criteria decision-making (MCDM) model based on complex circular intuitionistic fuzzy sets (CCIRIFSs) is presented in this study. This model combines the circular intuitionistic fuzzy set and complex numbers synergistically to address directional periodicity and two-dimensional uncertainties. On the other hand, to improve aggregation performance, two new operators are introduced, namely complex circular intuitionistic fuzzy dombi weighted averaging (CCIRIFDWA) and geometric (CCIRIFDWG). These operators, which have, of necessity, mathematical properties including idempotency, boundedness, and monotonicity, are built up of parameterized Dombi t-norms and t-conorms and thus meet the analytical requirements of rigour and flexibility in the processing of imprecise data. The case study presented in this study is a detailed analysis of sustainable agriculture, demonstrating that the given framework can be effective and useful for evaluating sustainability approaches across different criteria. The results established that greenhouse farming under a controlled environment was the best strategy, with scores of 0.874 for CCIRIFDWA and 0.861 for CCIRIFDWG, indicating the model's strong and flexible nature. Altogether, the given framework can be regarded as a novice, mathematically complete, and context-specific decision-making approach to cyclic, uncertain agricultural systems.

Keywords: MCDM; complex circular intuitionistic fuzzy sets; Dombi aggregation operators; sustainable agriculture; fuzzy methodologies; decision support systems

ARTICLE IN PRESS

1. Introduction

Sustainable agriculture is an issue that has sparked a global debate as the pressure on natural resources, food security, and environmental sustainability increases dramatically. The adoption of sustainable methods, such as accurate irrigation, crop rotation, organic farming, and the use of greenhouses to produce goods, requires that decision-makers consider economic feasibility and environmental protection. Such strategies are typically incompatible; their effects on the environment and on society are unpredictable. Consequently, the analysis and selection of sustainable agricultural practices under uncertainty constitute a multi-criteria decision-making (MCDM) problem. One will need to develop one to build a powerful analytical framework that can operate across different sustainability measures and address the uncertainty inherent in expert judgments. The decisions made regarding sustainability are becoming more complex, as agricultural planners face uncertain, circular environmental conditions. Conventional MCDM techniques do not account for such periodic uncertainty. To fill this gap, this paper proposes a fuzzy framework.

MCDM has become relevant, especially in complex environments where mutually supporting and ambiguous factors coexist and are not compatible. These cases involve dealing with several, and in most instances incompatible, criteria at the same time, as in environmental engineering, logistics, energy planning, and specifically in sustainable agriculture [1], [2], [3], [4]. Sustainable management of soil, water, biodiversity, and socio-economics must be integrated into the agricultural decision-making

ARTICLE IN PRESS

system, which will require applying complex MCDM models [5], [6]. Conventional decision-making methods tend to be confined to binary, crisp paradigms and fail to capture the fuzziness and subjectivity inherent in human decision-making. In the real world, issues, particularly in the sphere of sustainability, are hardly absolute. As [7] has indicated, this has brought to the fore the fuzziness, indecisiveness, and partial truth values of MCDM methods.

1.1. Literature Review

Over the years, various MCDM approaches have been used to address complex evaluation problems in systems concerned with sustainability. The classical models, including the Analytic Hierarchy Process (AHP) [8], the technique for order preference by similarity to ideal solution (TOPSIS) [9], and VIKOR [10], are well-known for their systematic and transparent evaluation processes. Nevertheless, the approaches are based on exact numerical information and linear preference hypotheses. As such, they can be restrictive in modeling subjective human pertinence and uncertainties that are typically apparent in the agricultural and environmental systems. These innovations led to the use of fuzzy set (FS) theory by Zadeh [11] to revolutionize decision analysis, as the representation of imprecise and linguistic data became possible. Later developed by Atanassov in his intuitionistic fuzzy sets (IFSs) [12], which included degrees of belonging, non-belonging, and hesitation, to provide a more human-like model of human thinking. Subsequently, interval-valued intuitionistic fuzzy models [13] and fuzzy-logic-based decision systems in agriculture [14] further improved the modeling of uncertainty in real-world scenarios.

In an effort to enhance the flexibility of aggregation, Seikh and Mandal [14] proposed intuitionistic fuzzy Dombi operators, and Khan et al. [15] subsequently enhanced them by incorporating lower and upper approximation mechanisms. Other more recent developments are complex and circular fuzzy models, e.g., the complex circular intuitionistic fuzzy Heronian mean model of dynamic air quality forecasting developed by Kong [16], and the Pythagorean fuzzy sets (PFS) [17] and generalized orthopair fuzzy sets [18], where the degree of belonging assignment can take place in a broader domain. In the meantime, neutrosophic systems [19], [20] have been designed to handle better indeterminacy and inconsistent information provided by experts. Several more advanced aggregation operators, such as the Bonferroni mean [21], the Einstein [22] and Hamacher [23] operators, have also been investigated to better reflect the dependence among criteria. However, these methods tend to support scalar or linear data formats and are therefore inadequate for describing cyclic relationships, directional uncertainty, and non-linear relationships between sustainability indicators. The above limitations serve as the impetus for the present research, which proposes a complex circular intuitionistic fuzzy (CCIRIF) framework that incorporates two-dimensional, cyclic, and parameterized Dombi aggregation operators to provide a more flexible and realistic decision-making model for sustainable agriculture. Archimedean Heronian mean operators were proposed by Ali et al. [24], which combine complex intuitionistic fuzzy sets (CIFS), which have a better performance in aggregation with MCDM problems.

Among them, Dombi aggregation operators stand out, particularly for their parameterized format, which allows varying the degree of compensation between decision attributes and thereby approximating human reasoning patterns more closely [25], [26]. CIRIFSs have also recently emerged as a means to represent both directional and value-based uncertainty [27]. In contrast to the usual fuzzy sets or intuitionistic sets, CIRIFSs are circularly represented, making them appropriate for cyclic phenomena such as seasonal crops, periodic rainfall, pest infestations, and closed supply chains [28], [29], [30]. Data in CIRIFSs are represented as angular and radial points on a unit circle, thereby offering a more intuitive way to represent hesitancy and conflict between criteria [31]. The sets provide an important benefit of multidimensional measurement, including sustainable agriculture, where indicators are usually recurrent and integrated across time [32]. Innovative and sustainable decision-making systems are also becoming relevant to a fuzzy-aggregation approach. Riaz and Farid [33] suggested using linear Diophantine fuzzy soft-max operators to minimize the efficiency of the green supply chain and to address supplier selection uncertainty. According to Saha et al. [34], Archimedean-Dombi operators based on similarity would be more effective in clustering imprecise data by using

ARTICLE IN PRESS

fuzzy similarity operated classification. Sugeno-Weber interval value spherical fuzzy models on the other hand were used by Sarfarz and Bozanic [35] to quantify recycled water and proved their abilities in terms of environmental analysis. The amalgamation of these works allows adding to the use of fuzzy logic in more reliable and sustainable decisions. Recent research has indicated the efficiency of Dombi-based fuzzy models in the context of uncertainty in different areas.

On the one hand, T-spherical fuzzy information was used by Khan and Rahman [36] to evaluate customer behavior in the automotive industry, and on the other hand, Ahmad [37] enhanced image segmentation using Einstein-Dombi operators. Additionally, Jana and Pal [38] used interval-valued picture-fuzzy Dombi operators to select industrial funds, highlighting the model's strength in complex decision-making environments.

It is worth noting that the concepts of the complex and CIRIF have already been investigated in the literature. For example, Ali et al. [24] developed Archimedean Heronian mean operators based on intricate IFSs and illustrated their use in decision-making. Although the current research is based on this premise, it expands on the idea by incorporating a circular representation into the Dombi aggregation framework, enabling the modelling of cyclic and directional uncertainties, which are especially pertinent in agricultural sustainability assessment. The more recent research has improved the fuzzy decision-making method, Khan et al. [39], who came up with a fractional fuzzy similarity model to deal with obstacles to green supply chain management, and Khan et al. [40], who invented a quasi-rung orthopair fuzzy aggregation model to make the best choice in selecting a vehicle, both improving the accuracy of the decision when faced with uncertainty. Fuzzy and complex decision-making models have been very important for managing uncertainty across various fields. Garg and Rehman [41] proposed a bipolar complex fuzzy group decision-making algorithm to assess hepatitis risk using sine-trigonometric functions, thereby improving the quality of medical decisions. Complex q-rung orthopair fuzzy Muirhead mean models were suggested by Hussein et al. [42] to make complex decisions under complex uncertainty. In the same way, Rehman et al. [43] considered uncertainty in battery energy storage assessment using a complex hesitant fuzzy multi-criteria decision-making approach with the Einstein operator, whereas Waqas et al. [45] prioritised geothermal energy systems using a hesitant bipolar fuzzy approach with Dombi aggregation for the industrial sector.

ARTICLE IN PRESS

1.2. Research Objectives

The primary objective of this research is to develop an integrated decision-making model that can effectively address uncertainties, indecisiveness, and periodicity in sustainability analysis. In particular, the study will seek to:

i. Develop a CCIRIF MCDM model, which considers both angular and two-dimensional uncertainty.
ii. Formulate and evaluate two new Dombi-based aggregation operators, namely the CCIRIFDWA and the CCIRIFDWG, which allow non-linear aggregation of fuzzy information that is flexible.
iii. Assess the sustainable agriculture strategy using the proposed model and determine if it is accurate, interpretable, and performs better than traditional fuzzy MCDM methods.

1.3. Role And Advantages Of Dombi Aggregation Operators

Dombi aggregation operators are parametric non-linear operators that allow for the flexible modeling of the interaction between criteria [44]. These operators, lying between the extremes of conjunctive and disjunctive behavior, depend on the parameter. They benefit from compatibility with fuzzy systems, making them ideal for real-world applications that require soft trade-offs [45], [46]. When applied to the CCIRIFs, Dombi aggregation operators also improve decision accuracy. This kind of investigation sets two new operators.

- Complex Circular Intuitionistic Fuzzy Dombi Weighted Averaging (CCIRIFDWA)
- Complex Circular Intuitionistic Fuzzy Dombi Weighted Geometric (CCIRIFDWG)

ARTICLE IN PRESS

These operators maintain angular hesitation and exhibit non-linear compensatory behavior, which is vital in assessing sustainability strategies in ambiguous situations [47], [48], [49].

1.4. Research Gap

The existing models of fuzzy MCDM are mostly linear or one-dimensional and cannot sufficiently capture the circularity of environmental phenomena or the complex interdependencies among the criteria of sustainability. Also, the available aggregation operators do not have the parametric flexibility to simulate human-like reasoning under uncertainty. This study bridges these gaps by combining circular intuitionistic and fuzzy modeling with Dombi parametric operators to provide a new approach that is more precise, flexible, and realistic in analyzing sustainable agricultural decisions. Most recent developments in fuzzy MCDM have introduced new methods for handling uncertainty and expert subjectivity. For example, Cakir and Tas [50] developed a circular intuitionistic fuzzy decision-making model. On the same note, Khan et al. [52] proposed a group decision-making framework based on circular intuitionistic fuzzy preference relations. These models are effective for directional uncertainty but are restricted to computational and engineering applications. In addition, Alkan and Kahraman [51] and Cakir et al. [52] have introduced multi-criteria decision-making models in an intuitionistic and a circular fuzzy environment, respectively. Still, they lack a cyclical pattern of the environment and the interdependence of sustainability indicators characteristic of the agricultural system.

The current fuzzy MCDM models also suffer from the limitation of linear or one-dimensional aggregation models, which lack the parametric capability to approximate human reasoning under uncertainty. As a result, they do not represent circular environmental dynamics and interdependence sustainability criteria that affect agricultural productivity. To fill these gaps, the current study combines circular intuitionistic fuzzy modeling with Dombi parametric aggregation to provide a generalized, flexible framework capable of addressing cyclical, directional, and uncertain assessments in agricultural sustainability.

1.5. Main Contributions Of The Present Study

The primary contributions of this study are as follows:

1. The new MCDM framework is based on CCIFSs and improved uncertainty modeling.
2. Blending of two new CCIRIFDWA and CCIRIFDWG to store non-linear and circular aggregates of data.
3. The proposed framework was applied to a hypothetical case study, in which five sustainable agricultural strategies were analyzed using five key criteria.
4. Comparison of the proposed method with classical models to demonstrate the flexibility, robustness, and interpretability of the proposed method.
5. The capability to support making expert decisions in situations of uncertainty, and being cyclical, such as in sustainability and agricultural planning.

1.6. Organization of Article

This paper is organized to provide a clear picture of the proposed framework. In Section 2, the mathematical preliminaries of complex circular intuitionistic fuzzy sets (CCIRIFSs) and Dombi operators are stated. In Section 3, two novel aggregation operators are presented along with their properties. In Section 4, the proposed MCDM framework is introduced. Section 5 uses the framework on a case study of sustainable agriculture. Section 6 presents a comparative analysis of current practices, and Section 7 concludes the study by outlining further areas of research. Figure 1 presents a comprehensive overview of our proposed study.

ARTICLE IN PRESS

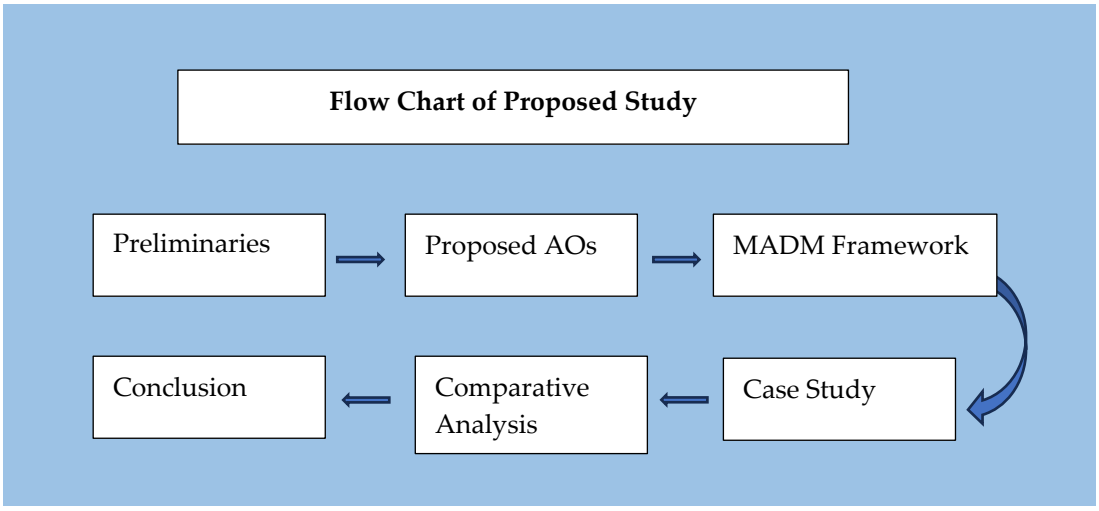


Figure 1. Pictorial overview of proposed work.

2. Preliminaries

To demonstrate the truth and significance of the given theory, this section provides a brief overview of some current methods, including Dombi t-norm (DTN), Dombi t-conorm (DTCN), and FS, IFS, CFS, CIFS, and CIRIFSs, applied to a universal set. β_ε

Definition 1. [11] Let β_ε be the universe of discurse, A fuzzy set $\mathfrak{H}$ is characterize as:

$$\mathfrak{H}(x) = (x, \text{K}(x) \,|x \in \beta_\varepsilon)$$

Where $\text{K}(x)\rightarrow[0, 1]$ presents the degree of belonging. The closer $\text{K}(x)$ is to 1, the more strongly x belongs to the set.

Example 1. Consider soil moisture as the basis for irrigation.

$$\mathfrak{H} = \{Dry, Moderate, Wet\}$$

An expert assigns

$$\mathfrak{H} = \{0.1, 0.8, 0.3\}$$

This demonstrates that moderately moist soil is the most suitable for irrigation (a suitability of 0.8), indicating that the fuzzy sets represent partial membership rather than binary states.

Defintion 2.[12] Let β_ε be the universe of discurse, A intuitionistic fuzzy set $\mathfrak{H}$ is characterize as:

$$\mathfrak{H}(x) = (x, \text{K}(x), \text{M}(x)|x \in \beta_\varepsilon)$$

Where $\text{K}(x)\rightarrow[0, 1]$ presents the degree of belonging, $\text{M}(x)\rightarrow[0, 1]$ presents the degree of non-belonging, with the condition $0 \leq \text{K}(x) + \text{M}(x) \leq 1$. The hesitancy degree of Intuitionistic fuzzy numbers is defined as:

$$\mathcal{H}(x) = \big(1 - \text{K}(x) - \text{M}(x)\big)$$

Example 2. A decision-maker evaluates that a type of fertilizer is environmentally sustainable. When $\text{K}(x) = 0.7$ (acts in favor of sustainability) and $\text{M}(x) = 0.2$ (acts against sustainability), the degree of hesitancy is:

$$\mathcal{H}(x) = (1 - 0.7 - 0.2) = 0.1$$

This implies that there is a 10 percent uncertainty in expert judgment, reflecting incomplete knowledge, a common feature of sustainability assessments.

Definition 3. [53] Let β_ε be the universe of discurse, A complex fuzzy set $\mathfrak{H}_c$ is characterize as:

$$\mathfrak{H}_c = \{x, \text{K}(x)e^{i\theta(x)}|x \in \beta_\varepsilon\}$$

Where $\text{K}(x)e^{i\theta(x)}$ is known as a complex valued degree of belonging, which further includes $\text{K}(x)\rightarrow[0, 1]$ as amplitude term and $\theta(x)\rightarrow[0, 2\pi]$ as a phase term with $i = \sqrt{-1}$.

Example 3. Assume that the patterns of rainfall impact on crop yield during a year. When the suitability of rainfall is $0.9e^{i\pi/3}$. This implies that the yield suitability of 0.9 and the maximum suitability at a phase occur in the third month of the cycle (spring). The phase component would be used to model behavior over time, as in classical fuzzy sets.

Definition 4. [54] A CIFS $\mathfrak{H}_c$ on a universal set β_ε is characterized as:

ARTICLE IN PRESS

$$\mathfrak{H}_c = \{x, \mathrm{K}(x), \mathrm{M}(x) | x \in \beta_\varepsilon\}$$

Where K, M are present, the complex degree of belonging and complex degree of non-belonging, respectively, and also $\mathrm{K}_p(x) = r_p(x)e^{i\theta_{r_p}(x)}$ and $\mathrm{M}_p(x) = s_p(x)e^{i\theta_{s_p}(x)}$. Where $0 \le r_p(x) + s_p(x) \le 1$ and $0 \le \theta_{r_p}(x) + \theta_{s_p}(x) \le 2\pi$ for all $x \in \beta_\varepsilon$.

Example 4. A farmer is comparing a planting method with the differing effects of the seasons.

Suppose $\mathrm{K}(x) = 0.8e^{ip\pi/4}$ (highly positive in early summer) and $\mathrm{M}(x) = 0.1e^{ip\pi/6}$ (a little negative in spring), the model will predict the extent and the time of usefulness of the method, providing a multidimensional explanation.

Definition 5 [27]**.**The CIRIFS (Circular intuitionistic fuzzy set) $\mathfrak{H}$ of some universal set β_ε is started as:

$$\mathfrak{H} = \left\{\left(\mathrm{K}_f(x), \mathrm{M}_g(x), \check{\mathrm{Y}}_o(x)\right) : x \in \beta_\varepsilon\right\}$$

Where the term $\mathrm{K}_f(x)$ presents the degree of belonging, $\mathrm{M}_g(x)$ presents the degree of non-belonging, such as $0 \le \mathrm{K}_f(x) + \mathrm{M}_g(x) \le 1$. Moreover $\check{\mathrm{Y}}_o(x) \in [0, 1]$ defines the radial distance on each element spread across the circular framework of $x \in \beta_\varepsilon$. Additionally, the hesitancy degree is shaped as $\mathcal{H}(x) = \left(1 - \mathrm{K}_f(x) - \mathrm{M}_g(x)\right)$. A triplet $\mathfrak{H}^j = \left(\mathrm{K}_f^j, \mathrm{M}_g^j, \check{\mathrm{Y}}_j\right), j = 1, 2, \ldots, Z$ is known as the CIRIFNs.

Example 5. Let $\beta_\varepsilon = (\mathfrak{H}_1, \mathfrak{H}_2, \mathfrak{H}_3)$ be the set of renewable energy strategies of sustainable agriculture:

$\mathfrak{H}_1$: Solar-powered irrigation,

$\mathfrak{H}_2$: Water pumping with the aid of wind,

$\mathfrak{H}_3$: Bioenergy agricultural waste.

The criterion used is the Evaluation criterion, Energy Reliability under Seasonal Variation.

The allotted CIRIFNs are:

$$\mathfrak{H}_1(x) = (0.8, 0.1, 0.8)$$
$$\mathfrak{H}_2(x) = (0.6, 0.3, 0.7)$$
$$\mathfrak{H}_3(x) = (0.7, 0.2, 0.8)$$

The hesitancy degrees:

$$\mathcal{H}_1(x) = (1 - 0.8 - 0.1) = 0.1$$
$$\mathcal{H}_2(x) = (1 - 0.6 - 0.3) = 0.1$$
$$\mathcal{H}_3(x) = (1 - 0.7 - 0.2) = 0.1$$

Interpretation:

The alternative $\mathfrak{H}_1(x)$ (solar-powered irrigation) has the largest belonging (0.8) and radial stability (0.9) , making it the most reliable against seasonal cycles. Conversely, $\mathfrak{H}_2(x)$ (wind-assisted pumping) bears non-belonging (0.3), indicating greater variation due to wind variability.

Definition 6 [55] Let $\check{\mathrm{R}}_1, \check{\mathrm{R}}_2 \in [0,1]$ be two positive real numbers with $\tau \ge 1$ denote a real-valued parameter. DTN or DTCN is formally defined by the following expression.

$$D_{TN}(\check{\mathrm{R}}_1, \check{\mathrm{R}}_2) = \frac{1}{1 + \left(\left(\frac{1 - \check{\mathrm{R}}_1}{\check{\mathrm{R}}_1}\right)^\tau + \left(\frac{1 - \check{\mathrm{R}}_2}{\check{\mathrm{R}}_2}\right)^\tau\right)^{\frac{1}{\tau}}} \in [0, 1]$$

$$D_{TCN}(\check{\mathrm{R}}_1, \check{\mathrm{R}}_2) = 1 - \frac{1}{1 + \left(\left(\frac{\check{\mathrm{R}}_1}{1 - \check{\mathrm{R}}_1}\right)^\tau + \left(\frac{\check{\mathrm{R}}_2}{1 - \check{\mathrm{R}}_2}\right)^\tau\right)^{\frac{1}{\tau}}} \in [0, 1]$$

Definition 7 [27]**.** For any two CIRIFNs $\mathfrak{H}^j = \left(\mathrm{K}_f^j, \mathrm{M}_g^j, \check{\mathrm{Y}}_j\right), j = 1, 2, \ldots, Z$, some basic operational laws are defined as:

1. $\mathfrak{H}^1 \cup_{max} \mathfrak{H}^2 = \left(\max\left(\mathrm{K}^1, \mathrm{K}^2\right), \min(\mathrm{M}^1, \mathrm{M}^2), \max(\check{\mathrm{Y}}_1, \check{\mathrm{Y}}_2)\right)$
2. $\mathfrak{H}^1 \cup_{min} \mathfrak{H}^2 = \left(\max(\mathrm{K}^1, \mathrm{K}^2), \min(\mathrm{M}^1, \mathrm{M}^2), \min(\check{\mathrm{Y}}_1, \check{\mathrm{Y}}_2)\right)$
3. $\mathfrak{H}^1 \cap_{max} \mathfrak{H}^2 = \left(\min\left(\mathrm{K}^1, \mathrm{K}^2\right), \max(\mathrm{M}^1, \mathrm{M}^2), \max(\check{\mathrm{Y}}_1, \check{\mathrm{Y}}_2)\right)$
4. $\mathfrak{H}^1 \cap_{min} \mathfrak{H}^2 = \left(\min\left(\mathrm{K}^1, \mathrm{K}^2\right), \max(\mathrm{M}^1, \mathrm{M}^2), \min(\check{\mathrm{Y}}_1, \check{\mathrm{Y}}_2)\right)$
5. $\mathfrak{H}^1 \oplus_t \mathfrak{H}^2 = \left(\mathrm{K}^1 + \mathrm{K}^2 - \mathrm{K}^1\mathrm{K}^2, (\mathrm{M}^1)(\mathrm{M}^2), \check{\mathrm{Y}}^1 + \check{\mathrm{Y}}^2 - \check{\mathrm{Y}}^1\check{\mathrm{Y}}^2\right)$
6. $\mathfrak{H}^1 \oplus_{tc} \mathfrak{H}^2 = \left(\mathrm{K}^1 + \mathrm{K}^2 - \mathrm{K}^1\mathrm{K}^2, (\mathrm{M}^1)(\mathrm{M}^2), (\check{\mathrm{Y}}^1)(\check{\mathrm{Y}}^2)\right)$
7. $\mathfrak{H}^1 \otimes_t \mathfrak{H}^2 = (\mathrm{K}^1\mathrm{K}^2, \mathrm{M}^1 + \mathrm{M}^2 - \mathrm{M}^1\mathrm{M}^2, \check{\mathrm{Y}}^1 + \check{\mathrm{Y}}^2 - \check{\mathrm{Y}}^1\check{\mathrm{Y}}^2)$
8. $\mathfrak{H}^1 \otimes_{tc} \mathfrak{H}^2 = (\mathrm{K}^1\mathrm{K}^2, \mathrm{M}^1 + \mathrm{M}^2 - \mathrm{M}^1\mathrm{M}^2, \check{\mathrm{Y}}^1\check{\mathrm{Y}}^2)$
9. $\propto_t \mathfrak{H} = \left(1 - \left(1 - \mathrm{K}_f\right)^\propto, \left(\mathrm{M}_g\right)^\propto, 1 - (1 - \check{\mathrm{Y}}_o)^\propto\right)$

ARTICLE IN PRESS

10. $\propto_{tc}\mathfrak{H} = \left(1-\left(1-\mathrm{K}_f\right)^{\propto}, \left(\mathrm{M}_g\right)^{\propto}, \left(\check{\Upsilon}_o\right)^{\propto}\right)$
11. $\mathfrak{H}^{\propto_t} = \left(\left(\mathrm{K}_f\right)^{\propto}, 1-\left(1-\mathrm{M}_g\right)^{\propto}, 1-\left(1-\check{\Upsilon}_o\right)^{\propto}\right)$
12. $\mathfrak{H}^{\propto_{tc}} = \left(\left(\mathrm{K}_f\right)^{\propto}, 1-\left(1-\mathrm{M}_g\right)^{\propto}, \left(\check{\Upsilon}_o\right)^{\propto}\right)$

Definition 8 [27]**.** Let $\mathfrak{H}^j = \left(\mathrm{K}_f^j, \mathrm{M}_g^j, \check{\Upsilon}_j\right), j = 1, 2$ be the CIRIFNs. The associated score function and accuracy function are given.

$$S(\mathfrak{H}^j) = \check{\Upsilon}_j * \left(\mathrm{K}_f^j - \mathrm{M}_g^j\right) \in [-1, 1]$$
$$H(\mathfrak{H}^j) = \check{\Upsilon}_j * \left(\mathrm{K}_f^j - \mathrm{M}_g^j\right) \in [0, 1]$$

2.1. Complex Circular Intuitionistic Fuzzy Set

The section presents the concept of CCIRIFSs and the fundamental laws governing their operation, including score and accuracy functions and algebraic structures. Moreover, we derive the operational rules for the Dombi combination, which combines two CCIRIFSs, and use them to develop a new type of aggregation operator.

Definition 9 A CCIRIFS $\mathfrak{H}$ defined on the universal set β_ε such as;

$$\mathfrak{H} = \left\{\left(\mathrm{K}_f(x), \mathrm{M}_g(x), \check{\Upsilon}_o(x)\right) : x \in \beta_\varepsilon\right\}$$

Where the term $\mathrm{K}_f(x) = \left(\mathrm{K}_f^{RP}(x), \mathrm{K}_f^{IP}(x)\right)$ presents the degree of belonging, $\mathrm{M}_g(x) = \left(\mathrm{M}_g^{RP}(x), \mathrm{M}_g^{IP}(x)\right)$ presents the degree of non-belonging, such as $0 \le \mathrm{K}_f^{RP}(x) + \mathrm{M}_g^{RP}(x) \le 1, 0 \le \mathrm{K}_f^{IP}(x) + \mathrm{M}_g^{IP}(x) \le 1$. Moreover $\check{\Upsilon}_o = \left(\check{\Upsilon}_o^{RP}, \check{\Upsilon}_o^{IP}\right) \in [0, 1] \times [0, 1]$ defines the radial distance on each element spread across the circular framework of $x \in \beta_\varepsilon$. The hesitancy degree is shaped as $\mathcal{H}(x) = \left(1 - \mathrm{K}_f^{RP} - \mathrm{M}_g^{RP}, 1 - \mathrm{K}_f^{IP} - \mathrm{M}_g^{IP}\right)$. A triplet $\mathfrak{H}^j = \left(\left(\mathrm{K}_{fPR}^j, \mathrm{K}_{fIR}^j\right), \left(\mathrm{M}_{gPR}^j, \mathrm{M}_{gIR}^j\right), \left(\check{\Upsilon}_j^{PR}, \check{\Upsilon}_j^{IR}\right)\right), j = 1, 2, \ldots, Z$ is known as the CCIRIFNs.

If we talk about the geometry of the CCIRIFS, a pair of complex numbers (*emebreship*,*etc*) in the complex plane, an image of each element of the universe. The phase angles and magnitudes of these functions describe the uncertainty and directional behavior, respectively. It is important to note that these values must lie in or on the specified circles centered at a predefined radius when plotting in the complex plane. The radius controls the permissible variation, while the phase term controls the phase-dependent information. As Atanassov provide the geometry of the circular IFS [27].

Example 6. Consider that there exists a set of sustainable agricultural strategies β_ε that is universally applicable

$$\beta_\varepsilon = (\mathfrak{H}_1, \mathfrak{H}_2, \mathfrak{H}_3)$$

$\mathfrak{H}_1$: Precision irrigation
$\mathfrak{H}_2$: Crop rotation
$\mathfrak{H}_3$: The use of organic fertilizers

Regarding the Water Resource Efficiency criterion, each strategy is rated using CCIRIF numbers within the established framework.

$$\mathfrak{H} = \left\{\left(\mathrm{K}_f(x), \mathrm{M}_g(x), \check{\Upsilon}_o(x)\right) : x \in \beta_\varepsilon\right\}$$
$$\mathcal{H}(\mathfrak{H}_1) = \left((0.7, 0.2), (0.2, 0.1), (0.8, 0.3)\right)$$
$$\mathcal{H}(\mathfrak{H}_2) = \left((0.6, 0.3), (0.3, 0.2), (0.7, 0.4)\right)$$
$$\mathcal{H}(\mathfrak{H}_2) = \left((0.8, 0.1), (0.1, 0.1), (0.9, 0.2)\right)$$

For every strategy, the hesitancy degree is calculated as

$$\mathcal{H}(\mathfrak{H}_1) = (1 - 0.7 - 0.2, 1 - 0.2 - 0.1) = (0.1, 0.7)$$
$$\mathcal{H}(\mathfrak{H}_2) = (1 - 0.6 - 0.3, 1 - 0.3 - 0.2) = (0.1, 0.5)$$
$$\mathcal{H}(\mathfrak{H}_3) = (1 - 0.8 - 0.1, 1 - 0.1 - 0.1) = (0.1, 0.8)$$

The findings indicate that $\mathfrak{H}_3$ (organic fertilizer application) possesses the most significant level of circular consistency with a lesser degree of belonging and a steady radial distance, implying that it is most effective in cyclic environmental factors.

Definition 10. For any two CCIRIFNs $\mathfrak{H}^j = \left(\mathrm{K}_f^j, \mathrm{M}_g^j, \check{\Upsilon}_j\right), j = 1, 2, \ldots, Z$, some fundamental operational laws are defined as:

ARTICLE IN PRESS

ARTICLE IN PRESS

1. $\mathfrak{H}^1 \cup_{max} \mathfrak{H}^2 = \left(\begin{pmatrix} \max\left(\mathrm{K}^1_{fRP}, \mathrm{K}^2_{fRP}\right), \\ \max\left(\mathrm{K}^1_{fIP}, \mathrm{K}^2_{fIP}\right) \end{pmatrix}, \begin{pmatrix} \min\left(\mathrm{M}^1_{jRP}, \mathrm{M}^2_{jRP}\right), \\ \min\left(\mathrm{M}^1_{jIP}, \mathrm{M}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \max\left(\check{\mathrm{Y}}^{RP}_1, \check{\mathrm{Y}}^{RP}_2\right), \\ \max\left(\check{\mathrm{Y}}^{IP}_1, \check{\mathrm{Y}}^{IP}_2\right) \end{pmatrix} \right)$

2. $\mathfrak{H}^1 \cup_{min} \mathfrak{H}^2 = \left(\begin{pmatrix} \max\left(\mathrm{K}^1_{fRP}, \mathrm{K}^2_{fRP}\right), \\ \max\left(\mathrm{K}^1_{fIP}, \mathrm{K}^2_{fIP}\right) \end{pmatrix}, \begin{pmatrix} \min\left(\mathrm{M}^1_{jRP}, \mathrm{M}^2_{jRP}\right), \\ \min\left(\mathrm{M}^1_{jIP}, \mathrm{M}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \min\left(\check{\mathrm{Y}}^{RP}_1, \check{\mathrm{Y}}^{RP}_2\right), \\ \min\left(\check{\mathrm{Y}}^{IP}_1, \check{\mathrm{Y}}^{IP}_2\right) \end{pmatrix} \right)$

3. $\mathfrak{H}^1 \cap_{max} \mathfrak{H}^2 = \left(\begin{pmatrix} \min\left(\mathrm{K}^1_{jRP}, \mathrm{K}^2_{jRP}\right), \\ \min\left(\mathrm{K}^1_{jIP}, \mathrm{K}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \max\left(\mathrm{M}^1_{jRP}, \mathrm{M}^2_{jRP}\right), \\ \max\left(\mathrm{M}^1_{jIP}, \mathrm{M}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \max\left(\check{\mathrm{Y}}^{RP}_1, \check{\mathrm{Y}}^{RP}_2\right), \\ \max\left(\check{\mathrm{Y}}^{IP}_1, \check{\mathrm{Y}}^{IP}_2\right) \end{pmatrix} \right)$

4. $\mathfrak{H}^1 \cap_{min} \mathfrak{H}^2 = \left(\begin{pmatrix} \min\left(\mathrm{K}^1_{jRP}, \mathrm{K}^2_{jRP}\right), \\ \min\left(\mathrm{K}^1_{jIP}, \mathrm{K}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \max\left(\mathrm{M}^1_{jRP}, \mathrm{M}^2_{jRP}\right), \\ \max\left(\mathrm{M}^1_{jIP}, \mathrm{M}^2_{jIP}\right) \end{pmatrix}, \begin{pmatrix} \min\left(\check{\mathrm{Y}}^{RP}_1, \check{\mathrm{Y}}^{RP}_2\right), \\ \min\left(\check{\mathrm{Y}}^{IP}_1, \check{\mathrm{Y}}^{IP}_2\right) \end{pmatrix} \right)$

5. $\mathfrak{H}^1 \oplus_t \mathfrak{H}^2 = \left(\begin{pmatrix} \mathrm{K}^1_{jRP} + \mathrm{K}^2_{jRP} - \mathrm{K}^1_{jRP}\mathrm{K}^2_{jRP}, \\ \mathrm{K}^1_{jRP} + \mathrm{K}^2_{jRP} - \mathrm{K}^1_{jRP}\mathrm{K}^2_{jRP} \end{pmatrix}, \left(\left(\mathrm{M}^1_{jRP}\right)\left(\mathrm{M}^2_{jRP}\right), \left(\mathrm{M}^1_{jIP}\right)\left(\mathrm{M}^2_{jIP}\right) \right), \begin{pmatrix} \check{\mathrm{Y}}^1_{RP} + \check{\mathrm{Y}}^2_{RP} - \check{\mathrm{Y}}^1_{RP}\check{\mathrm{Y}}^2_{RP}, \\ \check{\mathrm{Y}}^1_{RP} + \check{\mathrm{Y}}^2_{RP} - \check{\mathrm{Y}}^1_{RP}\check{\mathrm{Y}}^2_{RP} \end{pmatrix} \right)$

6. $\mathfrak{H}^1 \oplus_{tc} \mathfrak{H}^2 = \left(\begin{pmatrix} \mathrm{K}^1_{jRP} + \mathrm{K}^2_{jRP} - \mathrm{K}^1_{jRP}\mathrm{K}^2_{jRP}, \\ \mathrm{K}^1_{jRP} + \mathrm{K}^2_{jRP} - \mathrm{K}^1_{jRP}\mathrm{K}^2_{jRP} \end{pmatrix}, \left(\left(\mathrm{M}^1_{jRP}\right)\left(\mathrm{M}^2_{jRP}\right), \left(\mathrm{M}^1_{jIP}\right)\left(\mathrm{M}^2_{jIP}\right) \right) \right.$
$\left. , \left(\left(\check{\mathrm{Y}}^1_{RP}\right)\left(\check{\mathrm{Y}}^2_{RP}\right), \left(\check{\mathrm{Y}}^1_{IP}\right)\left(\check{\mathrm{Y}}^2_{IP}\right) \right) \right)$

7. $\mathfrak{H}^1 \otimes_t \mathfrak{H}^2 = \left(\begin{pmatrix} \mathrm{K}^1_{RP}\mathrm{K}^2_{RP}, \mathrm{K}^1_{IP}\mathrm{K}^2_{IP} \end{pmatrix}, \begin{pmatrix} \mathrm{M}^1_{RP} + \mathrm{M}^2_{RP} - \mathrm{M}^1_{RP}\mathrm{M}^2_{RP}, \\ \mathrm{M}^1_{IP} + \mathrm{M}^2_{IP} - \mathrm{M}^1_{IP}\mathrm{M}^2_{IP} \end{pmatrix}, \begin{pmatrix} \check{\mathrm{Y}}^1_{RP} + \check{\mathrm{Y}}^2_{RP} - \check{\mathrm{Y}}^1_{RP}\check{\mathrm{Y}}^2_{RP}, \\ \check{\mathrm{Y}}^1_{IP} + \check{\mathrm{Y}}^2_{IP} - \check{\mathrm{Y}}^1_{IP}\check{\mathrm{Y}}^2_{IP} \end{pmatrix} \right)$

8. $\mathfrak{H}^1 \otimes_{tc} \mathfrak{H}^2 = \left(\begin{pmatrix} \mathrm{K}^1_{RP}\mathrm{K}^2_{RP}, \mathrm{K}^1_{IP}\mathrm{K}^2_{IP} \end{pmatrix}, \begin{pmatrix} \mathrm{M}^1_{RP} + \mathrm{M}^2_{RP} - \mathrm{M}^1_{RP}\mathrm{M}^2_{RP}, \\ \mathrm{M}^1_{IP} + \mathrm{M}^2_{IP} - \mathrm{M}^1_{IP}\mathrm{M}^2_{IP} \end{pmatrix}, \begin{pmatrix} \check{\mathrm{Y}}^1_{RP}\check{\mathrm{Y}}^2_{RP}, \\ \check{\mathrm{Y}}^1_{IP}\check{\mathrm{Y}}^2_{IP} \end{pmatrix} \right)$

9. $\propto \mathfrak{H} = \left(\begin{pmatrix} 1 - \left(1 - \mathrm{K}_{fRP}\right)^{\propto}, \\ 1 - \left(1 - \mathrm{K}_{fIP}\right)^{\propto} \end{pmatrix}, \left(\left(\mathrm{M}_{gRP}\right)^{\propto}, \left(\mathrm{M}_{gIP}\right)^{\propto} \right), \right.$
$\left. \left(\left(1 - \left(1 - \check{\mathrm{Y}}_{oRP}\right)^{\propto}\right), 1 - \left(1 - \check{\mathrm{Y}}_{oIP}\right)^{\propto} \right) \right)$

10. $\propto \mathfrak{H} = \left(\begin{pmatrix} 1 - \left(1 - \mathrm{K}_{fRP}\right)^{\propto}, \\ 1 - \left(1 - \mathrm{K}_{fIP}\right)^{\propto} \end{pmatrix}, \right.$
$\left(\left(\mathrm{M}_{gRP}\right)^{\propto}, \left(\mathrm{M}_{gIP}\right)^{\propto} \right),$
$\left. \left(\check{\mathrm{Y}}_{oRP}\right)^{\propto}, \left(\check{\mathrm{Y}}_{oIP}\right)^{\propto} \right)$

11. $\mathfrak{H}^{\propto} = \left(\begin{pmatrix} \left(\mathrm{K}_{fRP}\right)^{\propto}, \\ \left(\mathrm{K}_{fIP}\right)^{\propto} \end{pmatrix}, \begin{pmatrix} 1 - \left(1 - \mathrm{M}_{gRP}\right)^{\propto}, \\ 1 - \left(1 - \mathrm{M}_{gIP}\right)^{\propto} \end{pmatrix}, \right.$
$\left. \begin{pmatrix} 1 - (1 - \check{\mathrm{Y}}_{oRP})^{\propto}, \\ 1 - (1 - \check{\mathrm{Y}}_{oIP})^{\propto} \end{pmatrix} \right)$

12. $\mathfrak{H}^{\propto} = \left(\left(\left(\mathrm{K}_{fRP}\right)^{\propto}, \left(\mathrm{K}_{fIP}\right)^{\propto} \right), \begin{pmatrix} 1 - \left(1 - \mathrm{M}_{gRP}\right)^{\propto}, \\ 1 - \left(1 - \mathrm{M}_{gIP}\right)^{\propto} \end{pmatrix}, \left(\left(\check{\mathrm{Y}}_{oRP}\right)^{\propto}, \left(\check{\mathrm{Y}}_{oIP}\right)^{\propto} \right) \right)$

Definition 11: Consider $\mathfrak{H}^j = \left(\mathrm{K}^j_f, \mathrm{M}^j_g, \check{\mathrm{Y}}_j\right), j = 1, 2$ be the two CCIRIFNs, the mathematical form of the score and accuracy function is given as.

$$S(\mathfrak{H}^j) = \left(\frac{\check{\mathrm{Y}}_{jRP} + \check{\mathrm{Y}}_{jIP}}{2}\right) * \left(\frac{\mathrm{K}_{jRP} - \mathrm{M}_{jRP} + \mathrm{K}_{jIP} - \mathrm{M}_{jIP}}{2}\right) \in [0, 1]$$

$$H(\mathfrak{H}^j) = \left(\frac{\check{\mathrm{Y}}_{jRP} + \check{\mathrm{Y}}_{jIP}}{2}\right) * \left(\frac{\mathrm{K}_{jRP} + \mathrm{M}_{jRP} + \mathrm{K}_{jIP} + \mathrm{M}_{jIP}}{2}\right) \in [0, 1]$$

Definition 12: Consider $\mathfrak{H}^j = \left(\mathrm{K}^j_f, \mathrm{M}^j_g, \check{\mathrm{Y}}_j\right), j = 1, 2$ be the two CCIRIFNs, then Dombi aggregation operators are defined as

$$\mathfrak{H}^1 \oplus_t \mathfrak{H}^2 =$$

ARTICLE IN PRESS

ARTICLE IN PRESS

$$\left(\left(\begin{array}{c}1-\frac{1}{1+\left(\left(\frac{\mathrm{K}^1_{jRP}}{1-\mathrm{K}^1_{jRP}}\right)^{\tau}+\left(\frac{\mathrm{K}^2_{jRP}}{1-\mathrm{K}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ 1-\frac{1}{1+\left(\left(\frac{\mathrm{K}^1_{jIP}}{1-\mathrm{K}^1_{jIP}}\right)^{\tau}+\left(\frac{\mathrm{K}^2_{jIP}}{1-\mathrm{K}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right),\left(\begin{array}{c}\frac{1}{1+\left(\left(\frac{1-\mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}}\right)^{\tau}+\left(\frac{1-\mathrm{M}^2_{gRP}}{\mathrm{M}^2_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ \frac{1}{1+\left(\left(\frac{1-\mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}}\right)^{\tau}+\left(\frac{1-\mathrm{M}^2_{gIP}}{\mathrm{M}^2_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

$$\left(\left(\begin{array}{c}1-\frac{1}{1+\left(\left(\frac{\breve{\mathrm{Y}}^1_{jRP}}{1-\breve{\mathrm{Y}}^1_{jRP}}\right)^{\tau}+\left(\frac{\breve{\mathrm{Y}}^2_{jRP}}{1-\breve{\mathrm{Y}}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ 1-\frac{1}{1+\left(\left(\frac{\breve{\mathrm{Y}}^1_{jIP}}{1-\breve{\mathrm{Y}}^1_{jIP}}\right)^{\tau}+\left(\frac{\breve{\mathrm{Y}}^2_{jIP}}{1-\breve{\mathrm{Y}}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

$$\mathfrak{H}^1\oplus_{tc}\mathfrak{H}^2=$$

$$\left(\left(\begin{array}{c}1-\frac{1}{1+\left(\left(\frac{\mathrm{K}^1_{jRP}}{1-\mathrm{K}^1_{jRP}}\right)^{\tau}+\left(\frac{\mathrm{K}^2_{jRP}}{1-\mathrm{K}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ 1-\frac{1}{1+\left(\left(\frac{\mathrm{K}^1_{jIP}}{1-\mathrm{K}^1_{jIP}}\right)^{\tau}+\left(\frac{\mathrm{K}^2_{jIP}}{1-\mathrm{K}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right),\left(\begin{array}{c}\frac{1}{1+\left(\left(\frac{1-\mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}}\right)^{\tau}+\left(\frac{1-\mathrm{M}^2_{gRP}}{\mathrm{M}^2_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ \frac{1}{1+\left(\left(\frac{1-\mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}}\right)^{\tau}+\left(\frac{1-\mathrm{M}^2_{gIP}}{\mathrm{M}^2_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

$$\left(\begin{array}{c}\frac{1}{1+\left(\left(\frac{1-\breve{\mathrm{Y}}^1_{jRP}}{\breve{\mathrm{Y}}^1_{jRP}}\right)^{\tau}+\left(\frac{1-\breve{\mathrm{Y}}^2_{jRP}}{\breve{\mathrm{Y}}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ \frac{1}{1+\left(\left(\frac{1-\breve{\mathrm{Y}}^1_{jIP}}{\breve{\mathrm{Y}}^1_{jIP}}\right)^{\tau}+\left(\frac{1-\breve{\mathrm{Y}}^2_{jIP}}{\breve{\mathrm{Y}}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

$$\mathfrak{H}^1\otimes_t\mathfrak{H}^2=\left(\left(\begin{array}{c}\frac{1}{1+\left(\left(\frac{1-\mathrm{K}^1_{jRP}}{\mathrm{K}^1_{jRP}}\right)^{\tau}+\left(\frac{1-\mathrm{K}^2_{jRP}}{\mathrm{K}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ \frac{1}{1+\left(\left(\frac{1-\mathrm{K}^1_{jIP}}{\mathrm{K}^1_{jIP}}\right)^{\tau}+\left(\frac{1-\mathrm{K}^2_{jIP}}{\mathrm{K}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right),\left(\begin{array}{c}1-\frac{1}{1+\left(\left(\frac{\mathrm{M}^1_{gIP}}{1-\mathrm{M}^1_{gIP}}\right)^{\tau}+\left(\frac{\mathrm{M}^2_{gRP}}{1-\mathrm{M}^2_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ 1-\frac{1}{1+\left(\left(\frac{\mathrm{M}^1_{gRP}}{1-\mathrm{M}^1_{gRP}}\right)^{\tau}+\left(\frac{\mathrm{M}^2_{gIP}}{1-\mathrm{M}^2_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

$$\left(\begin{array}{c}\frac{1}{1+\left(\left(\frac{1-\breve{\mathrm{Y}}^1_{jRP}}{\breve{\mathrm{Y}}^1_{jRP}}\right)^{\tau}+\left(\frac{1-\breve{\mathrm{Y}}^2_{jRP}}{\breve{\mathrm{Y}}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}},\\ \frac{1}{1+\left(\left(\frac{1-\breve{\mathrm{Y}}^1_{jIP}}{\breve{\mathrm{Y}}^1_{jIP}}\right)^{\tau}+\left(\frac{1-\breve{\mathrm{Y}}^2_{jIP}}{\breve{\mathrm{Y}}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\end{array}\right)\right)$$

ARTICLE IN PRESS

ARTICLE IN PRESS

$$\mathfrak{H}^1 \otimes_{tc} \mathfrak{H}^2 = \left\langle \left(\frac{1}{1+\left(\left(\frac{1-\mathrm{K}^1_{jRP}}{\mathrm{K}^1_{jRP}}\right)^{\tau}+\left(\frac{1-\mathrm{K}^2_{jRP}}{\mathrm{K}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\left(\frac{1-\mathrm{K}^1_{jIP}}{\mathrm{K}^1_{jIP}}\right)^{\tau}+\left(\frac{1-\mathrm{K}^2_{jIP}}{\mathrm{K}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\left(\frac{\mathrm{M}^1_{gIP}}{1-\mathrm{M}^1_{gIP}}\right)^{\tau}+\left(\frac{\mathrm{M}^2_{gRP}}{1-\mathrm{M}^2_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\left(\frac{\mathrm{M}^1_{gRP}}{1-\mathrm{M}^1_{gRP}}\right)^{\tau}+\left(\frac{\mathrm{M}^2_{gIP}}{1-\mathrm{M}^2_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\left(\frac{\breve{\Upsilon}^1_{jRP}}{1-\breve{\Upsilon}^1_{jRP}}\right)^{\tau}+\left(\frac{\breve{\Upsilon}^2_{jRP}}{1-\breve{\Upsilon}^2_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\left(\frac{\breve{\Upsilon}^1_{jIP}}{1-\breve{\Upsilon}^1_{jIP}}\right)^{\tau}+\left(\frac{\breve{\Upsilon}^2_{jIP}}{1-\breve{\Upsilon}^2_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right) \right\rangle$$

$$\propto_t \mathfrak{H}^j = \left\langle \left(1-\frac{1}{1+\left(\propto_t\left(\frac{\mathrm{K}^1_{jRP}}{1-\mathrm{K}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_t\left(\frac{\mathrm{K}^1_{jIP}}{1-\mathrm{K}^1_{jiP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1+\left(\propto_t\left(\frac{1-\mathrm{M}^1_{grP}}{\mathrm{M}^1_{grP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_t\left(\frac{1-\mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\propto_t\left(\frac{\breve{\Upsilon}^1_{jRP}}{1-\breve{\Upsilon}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_t\left(\frac{\breve{\Upsilon}^1_{jiP}}{1-\breve{\Upsilon}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right) \right\rangle$$

$$\propto_{tc} \mathfrak{H}^j = \left\langle \left(1-\frac{1}{1+\left(\propto_{tc}\left(\frac{\mathrm{K}^1_{jRP}}{1-\mathrm{K}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_{tc}\left(\frac{\mathrm{K}^1_{jIP}}{1-\mathrm{K}^1_{jiP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1+\left(\propto_{tc}\left(\frac{1-\mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_{tc}\left(\frac{1-\mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1+\left(\propto_{tc}\left(\frac{1-\breve{\Upsilon}^1_{jRP}}{\breve{\Upsilon}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_{tc}\left(\frac{1-\breve{\Upsilon}^1_{jiP}}{\breve{\Upsilon}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right) \right\rangle$$

$$(\mathfrak{H})^{\propto_t} = \left\langle \left(\frac{1}{1+\left(\propto_t\left(\frac{1-\mathrm{K}^1_{jRP}}{\mathrm{K}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_t\left(\frac{1-\mathrm{K}^1_{jIP}}{\mathrm{K}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\propto_t\left(\frac{\mathrm{M}^1_{gRP}}{1-\mathrm{M}^1_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_t\left(\frac{\mathrm{M}^1_{gIP}}{1-\mathrm{M}^1_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1+\left(\propto_t\left(\frac{1-\breve{\Upsilon}^1_{jRP}}{\breve{\Upsilon}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_t\left(\frac{1-\breve{\Upsilon}^1_{jiP}}{\breve{\Upsilon}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \right) \right\rangle$$

ARTICLE IN PRESS

$$(\mathfrak{H})^{\propto_{tc}} = \left(\left(\frac{1}{1+\left(\propto_{tc} \left(\frac{1-\mathrm{K}^1_{jRP}}{\mathrm{K}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1+\left(\propto_{tc} \left(\frac{1-\mathrm{K}^1_{jIP}}{\mathrm{K}^1_{jIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\propto_{tc} \left(\frac{\mathrm{M}^1_{gRP}}{1-\mathrm{M}^1_{gRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_{tc} \left(\frac{\mathrm{M}^1_{gIP}}{1-\mathrm{M}^1_{gIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1-\frac{1}{1+\left(\propto_{tc} \left(\frac{\check{\mathrm{Y}}^1_{jRP}}{1-\check{\mathrm{Y}}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1-\frac{1}{1+\left(\propto_{tc} \left(\frac{\check{\mathrm{Y}}^1_{jIP}}{1-\check{\mathrm{Y}}^1_{jIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right)$$

Table 1 presents the complete details of variables and parameters.

Table 1. Detailed overview of the variables

Variables	Detailes
$\mathbb{R}_1, \mathbb{R}_2$	Real numbers
$\boldsymbol{\tau}$	Parameter
$\boldsymbol{D_{TN}}$	Dombi t-norm
$\boldsymbol{D_{TCN}}$	Dombi t-conorm
$\boldsymbol{\beta_{\varepsilon}}$	Universal set
$\mathrm{K}_f(\boldsymbol{x})$	Degree of belonging
$\mathrm{M}_g(\boldsymbol{x})$	Degree of Non-Belonging
$\check{\mathrm{Y}}_o(\boldsymbol{x})$	Radius
$\propto$	Scalar element
$\boldsymbol{S}(\mathfrak{H}^j)$	Score Function
$\boldsymbol{H}(\mathfrak{H}^j)$	Accuracy Function

3. Proposed Dombi Aggregation Operators for CCIRIFSs

To integrate the information using the Dombi t-norm and t-conorm framework, we proposed some weighted aggregators. The desirable properties of these operators are also mentioned. These include:

i. Complex Circular Intuitionistic Fuzzy (CCIRIF)
ii. CCIRIF Dombi weighted averaging (CCIRIFDWA)
iii. CCIRIF Dombi Weighted Geometric (CCIRIFDWG)

Definition 13. Consider $\mathfrak{H}^j = \left(\mathrm{K}^j_f, \mathrm{M}^j_g, \check{\mathrm{Y}}_j\right), j = 1, 2, \ldots, Z$, be the family of CCIRIFNs, then CCIRIFDWA is defined as:

$$CCIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \propto_1 \mathfrak{H}^1 \oplus \propto_2 \mathfrak{H}^2 \oplus \ldots \oplus \propto_Z \mathfrak{H}^Z = \oplus_{j=1}^{Z} \propto_j \mathfrak{H}^j$$

$$CCIRIFDWA_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \propto_1 \mathfrak{H}^1 \oplus \propto_2 \mathfrak{H}^2 \oplus \ldots \oplus \propto_Z \mathfrak{H}^Z = \oplus_{j=1}^{Z} \propto_j \mathfrak{H}^j$$

Where $\propto_j \in [0, 1]$ shows the weight vector (WV) with $\sum_{j=1}^{Z} \propto_j = 1$.

Theorem 1. Consider $\mathfrak{H}^j = \left(\mathrm{K}^j_f, \mathrm{M}^j_g, \check{\mathrm{Y}}_j\right), j = 1, 2, \ldots, Z$, be the family of CCIRIFNs. Then the aggregated outcomes of the above equation are again CCIRIFN, and are also provided as

$$CCIRIFDWA_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)$$
$$=\left\langle\begin{array}{c}\left(1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{jRP}^1}{1-\mathrm{K}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{jIP}^1}{1-\mathrm{K}_{jiP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\mathrm{M}_{gRP}^1}{\mathrm{M}_{gRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\mathrm{M}_{gIP}^1}{\mathrm{M}_{gIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\mathrm{Y}}_{jRP}^1}{1-\check{\mathrm{Y}}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\mathrm{Y}}_{jiP}^1}{1-\check{\mathrm{Y}}_{jIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right)\end{array}\right\rangle$$

$$CCIRIFDWA_{tc}(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)$$
$$=\left\langle\begin{array}{c}\left(1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{jRP}^1}{1-\mathrm{K}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{jIP}^1}{1-\mathrm{K}_{jiP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\mathrm{M}_{gRP}^1}{\mathrm{M}_{gRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\mathrm{M}_{gIP}^1}{\mathrm{M}_{gIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\check{\mathrm{Y}}_{gRP}^1}{\check{\mathrm{Y}}_{gRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{1-\check{\mathrm{Y}}_{giP}^1}{\check{\mathrm{Y}}_{gIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right)\end{array}\right\rangle$$

Proof: We intend to analyze the information by the principle of mathematical induction. By assuming $Z=2$.

$$\propto_1\mathfrak{H}^1=\left\langle\begin{array}{c}\left(1-\frac{1}{1+\left(\propto_1\left(\frac{\mathrm{K}_{jRP}^1}{1-\mathrm{K}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\propto_1\left(\frac{\mathrm{K}_{jIP}^1}{1-\mathrm{K}_{jiP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(\frac{1}{1+\left(\propto_1\left(\frac{1-\mathrm{M}_{gRP}^1}{\mathrm{M}_{gRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},\frac{1}{1+\left(\propto_1\left(\frac{1-\mathrm{M}_{gIP}^1}{\mathrm{M}_{gIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(1-\frac{1}{1+\left(\propto_1\left(\frac{\check{\mathrm{Y}}_{jRP}^1}{1-\check{\mathrm{Y}}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\propto_1\left(\frac{\check{\mathrm{Y}}_{jiP}^1}{1-\check{\mathrm{Y}}_{jIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right)\end{array}\right\rangle$$

$$\propto_2\mathfrak{H}^2=\left\langle\begin{array}{c}\left(1-\frac{1}{1+\left(\propto_2\left(\frac{\mathrm{K}_{jRP}^1}{1-\mathrm{K}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\propto_2\left(\frac{\mathrm{K}_{jIP}^1}{1-\mathrm{K}_{jiP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(\frac{1}{1+\left(\propto_2\left(\frac{1-\mathrm{M}_{gRP}^1}{\mathrm{M}_{gRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},\frac{1}{1+\left(\propto_2\left(\frac{1-\mathrm{M}_{gIP}^1}{\mathrm{M}_{gIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right),\\ \left(1-\frac{1}{1+\left(\propto_2\left(\frac{\check{\mathrm{Y}}_{jRP}^1}{1-\check{\mathrm{Y}}_{jRP}^1}\right)^\tau\right)^{\frac{1}{\tau}}},1-\frac{1}{1+\left(\propto_2\left(\frac{\check{\mathrm{Y}}_{jiP}^1}{1-\check{\mathrm{Y}}_{jIP}^1}\right)^\tau\right)^{\frac{1}{\tau}}}\right)\end{array}\right\rangle$$

Hence,

ARTICLE IN PRESS

ARTICLE IN PRESS

$$\propto_1 \mathfrak{H}^1 \oplus \propto_2 \mathfrak{H}^2 = \left\langle \left(\left(1 - \frac{1}{1 + \left(\propto_1 \left(\frac{\mathrm{K}^1_{jRP}}{1 - \mathrm{K}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\propto_1 \left(\frac{\mathrm{K}^1_{jIP}}{1 - \mathrm{K}^1_{jiP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1 + \left(\propto_1 \left(\frac{1 - \mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\propto_1 \left(\frac{1 - \mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1 - \frac{1}{1 + \left(\propto_1 \left(\frac{\check{\mathrm{Y}}^1_{jRP}}{1 - \check{\mathrm{Y}}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\propto_1 \left(\frac{\check{\mathrm{Y}}^1_{jiP}}{1 - \check{\mathrm{Y}}^1_{jIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right) \right\rangle$$

$$\oplus \left\langle \left(\left(1 - \frac{1}{1 + \left(\propto_2 \left(\frac{\mathrm{K}^1_{jRP}}{1 - \mathrm{K}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\propto_2 \left(\frac{\mathrm{K}^1_{jIP}}{1 - \mathrm{K}^1_{jiP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1 + \left(\propto_2 \left(\frac{1 - \mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\propto_2 \left(\frac{1 - \mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1 - \frac{1}{1 + \left(\propto_2 \left(\frac{\check{\mathrm{Y}}^1_{jRP}}{1 - \check{\mathrm{Y}}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\propto_2 \left(\frac{\check{\mathrm{Y}}^1_{jiP}}{1 - \check{\mathrm{Y}}^1_{jIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right) \right\rangle$$

$$= \left\langle \left(\left(1 - \frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{\mathrm{K}^1_{jRP}}{1 - \mathrm{K}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{\mathrm{K}^1_{jIP}}{1 - \mathrm{K}^1_{jiP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(\frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{1 - \mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{1 - \mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{\check{\mathrm{Y}}^1_{jRP}}{1 - \check{\mathrm{Y}}^1_{jRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{2} \propto_j \left(\frac{\check{\mathrm{Y}}^1_{jiP}}{1 - \check{\mathrm{Y}}^1_{jIP}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right) \right\rangle$$

For $Z = 2$, it is true, now we consider that for $Z = k$, it is assumed to be true, such as

Further, we calculate for $Z = k + 1$

$$CCIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \propto_1 \mathfrak{H}^1 \oplus \propto_2 \mathfrak{H}^2 \oplus \ldots \oplus \propto_{k+1} \mathfrak{H}^{k+1} = \oplus_{j=1}^{k} \propto_k \mathfrak{H}^k \oplus_{j=1}^{k+1} \propto_{k+1} \mathfrak{H}^{k+1}$$

$$CCIRIFDWA_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \propto_1 \mathfrak{H}^1 \oplus \propto_2 \mathfrak{H}^2 \oplus \ldots \oplus \propto_{k+1} \mathfrak{H}^{k+1} = \oplus_{j=1}^{k} \propto_k \mathfrak{H}^k \oplus_{j=1}^{k+1} \propto_{k+1} \mathfrak{H}^{k+1}$$

ARTICLE IN PRESS

$$= \left\langle \begin{array}{c} \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{\mathrm{K}_{jRP}^1}{1 - \mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{\mathrm{K}_{jIP}^1}{1 - \mathrm{K}_{jiP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(\frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{1 - \mathrm{M}_{gRP}^1}{\mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{1 - \mathrm{M}_{gIP}^1}{\mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jRP}^1}{1 - \breve{\mathrm{Y}}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jiP}^1}{1 - \breve{\mathrm{Y}}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \end{array} \right\rangle$$

$$\oplus \left\langle \begin{array}{c} \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\mathrm{K}_{jRP}}{1 - \mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\mathrm{K}_{jIP}}{1 - \mathrm{K}_{jiP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(\frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jRP}}{1 - \breve{\mathrm{Y}}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jiP}}{1 - \breve{\mathrm{Y}}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \end{array} \right\rangle$$

$$= \left\langle \begin{array}{c} \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\mathrm{K}_{jRP}}{1 - \mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\mathrm{K}_{jIP}}{1 - \mathrm{K}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(\frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jRP}}{1 - \breve{\mathrm{Y}}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{k+1} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jiP}}{1 - \breve{\mathrm{Y}}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \end{array} \right\rangle$$

Therefore, we successfully tested our desired outcome among all positive numbers.

Property 1. (Idempotency) If $\mathfrak{H}^j = \left(\left(\mathrm{K}_{fRP}^j, \mathrm{K}_{fIP}^j \right), \left(\mathrm{M}_{gRP}^j, \mathrm{M}_{gIP}^j \right), \left(\breve{\mathrm{Y}}_j^{RP} \right) \right), j = 1, 2, \ldots, Z$ thus.

$$CIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \mathfrak{H}$$

$$CIRIFDWA_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \mathfrak{H}$$

Proof: Consider that $\mathfrak{H}^1 = \mathfrak{H} = \left(\left(\mathrm{K}_{fRP}, \mathrm{K}_{fIP} \right), \left(\mathrm{M}_{gRP}, \mathrm{M}_{gIP} \right), \left(\breve{\mathrm{Y}}^{RP} \right) \right), j = 1, 2, \ldots, Z$

$CCIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z)$

$$= \left\langle \begin{array}{c} \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{K}_{jRP}}{1 - \mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{K}_{jIP}}{1 - \mathrm{K}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(\frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \\ \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jRP}}{1 - \breve{\mathrm{Y}}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\breve{\mathrm{Y}}_{jIP}}{1 - \breve{\mathrm{Y}}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \end{array} \right\rangle$$

ARTICLE IN PRESS

$$= \left(\begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{K}_{RP}}{1 - \mathrm{K}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{K}_{IP}}{1 - \mathrm{K}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix}, \begin{pmatrix} \dfrac{1}{1 + \left(\left(\dfrac{1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \dfrac{1}{1 + \left(\left(\dfrac{1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix}, \begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{\breve{\mathrm{Y}}_{RP}}{1 - \breve{\mathrm{Y}}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \dfrac{1}{1 + \left(\left(\dfrac{\breve{\mathrm{Y}}_{IP}}{1 - \breve{\mathrm{Y}}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix} \right), \sum_{j=1}^{Z} \propto_{j}$$

$$= \left(\begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{K}_{RP}}{1 - \mathrm{K}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{K}_{IP}}{1 - \mathrm{K}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix}, \begin{pmatrix} \dfrac{1}{1 + \left(\left(\dfrac{1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ \dfrac{1}{1 + \left(\left(\dfrac{1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix} \right)$$

$$\left(\begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{\breve{\mathrm{Y}}_{RP}}{1 - \breve{\mathrm{Y}}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\left(\dfrac{\breve{\mathrm{Y}}_{IP}}{1 - \breve{\mathrm{Y}}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix} \right)$$

$$= \left(\begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{1 - \breve{\mathrm{Y}}_{RP} + \mathrm{K}_{RP}}{1 - \mathrm{K}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\left(\dfrac{1 - \mathrm{K}_{IP} + \mathrm{K}_{IP}}{1 - \mathrm{K}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix}, \begin{pmatrix} \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{M}_{gRP} + 1 - \mathrm{M}_{gRP}}{\mathrm{M}_{gRP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ \dfrac{1}{1 + \left(\left(\dfrac{\mathrm{M}_{gIP} + 1 - \mathrm{M}_{gIP}}{\mathrm{M}_{gIP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix} \right)$$

$$\left(\begin{pmatrix} 1 - \dfrac{1}{1 + \left(\left(\dfrac{1 - \breve{\mathrm{Y}}_{RP} + \breve{\mathrm{Y}}_{RP}}{1 - \breve{\mathrm{Y}}_{RP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\left(\dfrac{1 - \breve{\mathrm{Y}}_{IP} + \breve{\mathrm{Y}}_{IP}}{1 - \breve{\mathrm{Y}}_{IP}^{1}} \right)^{\tau} \right)^{\frac{1}{\tau}}} \end{pmatrix} \right)$$

$$\left(\left(\mathrm{K}_{fRP}, \mathrm{K}_{fIP} \right), \left(\mathrm{M}_{gRP}, \mathrm{M}_{gIP} \right), \left(\breve{\mathrm{Y}}^{RP} \right) \right) = \mathfrak{H}$$

Property 2. (Monotonicity) If $\mathfrak{H}^{j} = \left(\mathrm{K}_{fIP}^{j}, \mathrm{M}_{gRP}^{j} \right), \mathfrak{H}^{**j} = \left(\mathrm{K}^{**j}_{fIP}, \mathrm{M}^{**j}_{gRP} \right)$ are CCIRIFNs such as $\mathrm{K}_{fIP}^{j} \leq \mathrm{K}^{**j}_{fIP}$ and $\mathrm{M}_{gRP}^{j} \geq \mathrm{M}^{**j}_{gRP}$, then

$$CCIRIFDWA_{t}(\mathfrak{H}^{1}, \mathfrak{H}^{2}, \ldots, \mathfrak{H}^{Z}) \leq CIRIFDWA_{t}(\mathfrak{H}^{**1}, \mathfrak{H}^{**2}, \ldots, \mathfrak{H}^{**Z})$$

$$CCIRIFDWA_{tc}(\mathfrak{H}^{1}, \mathfrak{H}^{2}, \ldots, \mathfrak{H}^{Z}) \leq CIRIFDWA_{t}(\mathfrak{H}^{**1}, \mathfrak{H}^{**2}, \ldots, \mathfrak{H}^{**Z})$$

Proof. As $\mathrm{K}_{fIP}^{j} \leq \mathrm{K}^{**j}_{fIP}$ then $1 - \mathrm{K}_{fRP}^{j} \geq 1 - \mathrm{K}^{**j}_{fRP}$.

$$-\frac{\mathrm{K}_{fRP}^{j}}{1 - \mathrm{K}_{fRP}^{j}} \geq -\frac{\mathrm{K}^{**j}_{fRP}}{1 - \mathrm{K}^{**j}_{fRP}}$$

$$\Rightarrow -\left(\sum_{j=1}^{Z} \propto_{j} \left(\frac{\mathrm{K}_{fRP}^{j}}{1 - \mathrm{K}_{fRP}^{j}} \right)^{\tau} \right)^{\frac{1}{\tau}} \geq -\left(\sum_{j=1}^{Z} \propto_{j} \left(\frac{\mathrm{K}^{**j}_{fRP}}{1 - \mathrm{K}^{**j}_{fRP}} \right)^{\tau} \right)^{\frac{1}{\tau}}$$

ARTICLE IN PRESS

$$\Rightarrow -\left(1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{fRP}^{j}}{1-\mathrm{K}_{fRP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}\right)\geq -\left(1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}^{**j}_{fRP}}{1-\mathrm{K}^{**j}_{fRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}\right)$$

$$\Rightarrow -\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{fRP}^{j}}{1-\mathrm{K}_{fRP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\leq -\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}^{**j}_{fRP}}{1-\mathrm{K}^{**j}_{fRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

$$\Rightarrow 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{fRP}^{j}}{1-\mathrm{K}_{fRP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\leq 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}^{**j}_{fRP}}{1-\mathrm{K}^{**j}_{fRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

Also

$$\mathrm{K}_{fIP}^{j}\leq \mathrm{K}^{**j}_{fIP}$$

$$\Rightarrow 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}_{fIP}^{j}}{1-\mathrm{K}_{fIP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\leq 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{K}^{**j}_{fIP}}{1-\mathrm{K}^{**j}_{fIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

Further, for non-belonging and radius, the following are evaluated as:

$$\mathrm{M}_{gRP}^{j}\geq \mathrm{M}^{**j}_{gRP}\Rightarrow \frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{M}_{gRP}^{j}}{1-\mathrm{M}_{gRP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\geq \frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{M}^{**j}_{gRP}}{1-\mathrm{M}^{**j}_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

$$\mathrm{M}_{gIP}^{j}\geq \mathrm{M}^{**j}_{gIP}\Rightarrow \frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{M}_{gIP}^{j}}{1-\mathrm{M}_{gIP}^{j}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\geq \frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\mathrm{M}^{**j}_{gIP}}{1-\mathrm{M}^{**j}_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

For radius,

$$\check{\Upsilon}^{RP}\leq \check{\Upsilon}^{RP}_{**}\Rightarrow 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\Upsilon}^{RP}}{1-\check{\Upsilon}^{RP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\leq 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\Upsilon}^{RP}_{**}}{1-\check{\Upsilon}^{\overline{RP}}_{**}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

$$\check{\Upsilon}^{IP}\leq \check{\Upsilon}^{IP}_{**}\Rightarrow 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\Upsilon}^{IP}}{1-\check{\Upsilon}^{IP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}\leq 1-\frac{1}{1+\left(\sum_{j=1}^{Z}\propto_j\left(\frac{\check{\Upsilon}^{IP}_{**}}{1-\check{\Upsilon}^{\overline{IP}}_{**}}\right)^{\tau}\right)^{\frac{1}{\tau}}}$$

Hence proved.

Property 3. (Boundedness) If $\mathfrak{H}^{-}=\begin{pmatrix}\left(min\mathrm{K}_{RP}^{j},min\mathrm{K}_{IP}^{j}\right),\left(max\mathrm{M}_{RP}^{j},max\mathrm{M}_{IP}^{j}\right),\\ \left(min\check{\Upsilon}_{j}^{RP},min\check{\Upsilon}_{j}^{IP}\right)\end{pmatrix}$ and $\mathfrak{H}^{+}=\begin{pmatrix}\left(max\mathrm{K}_{RP}^{j},max\mathrm{K}_{IP}^{j}\right),\left(min\mathrm{M}_{RP}^{j},min\mathrm{M}_{IP}^{j}\right),\\ \left(max\check{\Upsilon}_{j}^{RP},max\check{\Upsilon}_{j}^{IP}\right)\end{pmatrix}, j=1,2,\ldots,Z$ thus.

$$\mathfrak{H}^{-}\leq CCIRIFDWA_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq \mathfrak{H}^{+}$$
$$\mathfrak{H}^{-}\leq CCIRIFDWA_{tc}(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq \mathfrak{H}^{+}$$

Proof: By using the date in Property 1, we have Property 2 as

$$CCIRIFDWA_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq CCIRIFDWA_t(\mathfrak{H}^{+1},\mathfrak{H}^{+2},\ldots,\mathfrak{H}^{+Z})=\mathfrak{H}^{+}$$
$$CCIRIFDWA_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq CCIRIFDWA_t(\mathfrak{H}^{-1},\mathfrak{H}^{-2},\ldots,\mathfrak{H}^{-Z})=\mathfrak{H}^{-}$$

Hence

$$\mathfrak{H}^{-}\leq CCIRIFDWA_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq \mathfrak{H}^{+}$$
$$\mathfrak{H}^{-}\leq CCIRIFDWA_{tc}(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)\leq \mathfrak{H}^{+}$$

Definition 11. Consider $\mathfrak{H}^{j}=\left(\mathrm{K}_{f}^{j},\mathrm{M}_{g}^{j},\check{\Upsilon}_{j}\right), j=1,2,\ldots,Z,$ be the family of CCIRIFNs, then the CCIRIFDWG operator is defined as:

$$CCIRIFDWG_t(\mathfrak{H}^1,\mathfrak{H}^2,\ldots,\mathfrak{H}^Z)=(\mathfrak{H}^1)^{\propto_1}\otimes(\mathfrak{H}^2)^{\propto_2}\otimes\ldots\otimes(\mathfrak{H}^Z)^{\propto_Z}=\otimes_{j=1}^{Z}(\mathfrak{H}^j)^{\propto_j}$$

ARTICLE IN PRESS

$$CCIRIFDWG_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = (\mathfrak{H}^1)^{\propto_1} \otimes (\mathfrak{H}^2)^{\propto_2} \otimes \ldots \otimes (\mathfrak{H}^Z)^{\propto_Z} = \otimes_{j=1}^{Z} (\mathfrak{H}^j)^{\propto_j}$$

Where $\propto_j \in [0, 1]$ shows the WV with the condition $\sum_{j=1}^{Z} \propto_j = 1$.

Theorem 3. Consider $\mathfrak{H}^j = \left(\mathrm{K}_f^j, \mathrm{M}_g^j, \check{\Upsilon}_j\right), j = 1, 2, \ldots, Z$, be the family of CCIRIFNs. Then the aggregated outcomes of the above equation are again CCIRIFN, which is also provided.

$$CCIRIFDWG_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z)$$

$$= \left(\left(\frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{K}_{jRP}^1}{\mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{K}_{jIP}^1}{\mathrm{K}_{jiP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{M}_{gRP}^1}{1 - \mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{M}_{gIP}^1}{1 - \mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right.$$

$$\left. \left(\frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \check{\Upsilon}_{jRP}^1}{\check{\Upsilon}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \check{\Upsilon}_{jiP}^1}{\check{\Upsilon}_{jIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right)$$

$$CCIRIFDWG_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z)$$

$$= \left(\left(\frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{K}_{jRP}^1}{\mathrm{K}_{jRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{1 - \mathrm{K}_{jIP}^1}{\mathrm{K}_{jiP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right), \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{M}_{gRP}^1}{1 - \mathrm{M}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\mathrm{M}_{gIP}^1}{1 - \mathrm{M}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right.$$

$$\left. \left(1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\check{\Upsilon}_{gRP}^1}{1 - \check{\Upsilon}_{gRP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}}, 1 - \frac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\frac{\check{\Upsilon}_{giP}^1}{1 - \check{\Upsilon}_{gIP}^1} \right)^{\tau} \right)^{\frac{1}{\tau}}} \right) \right)$$

Proof: Omitted

Property 4. (Idempotency) If $\mathfrak{H}^j = \left(\left(\mathrm{K}_{fRP}^j, \mathrm{K}_{fIP}^j \right), \left(\mathrm{M}_{gRP}^j, \mathrm{M}_{gIP}^j \right), \left(\check{\Upsilon}_j^{RP} \right) \right), j = 1, 2, \ldots, Z$ thus.

$$CCIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \mathfrak{H}$$

$$CCIRIFDWA_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) = \mathfrak{H}$$

Property 5 (Monotonicity) If $\mathfrak{H}^j = \left(\mathrm{K}_{fIP}^j, \mathrm{M}_{gRP}^j \right), \mathfrak{H}^{**j} = \left(\mathrm{K}^{**j}_{fIP}, \mathrm{M}^{**j}_{gRP} \right)$ are CCIRIFNs such as $\mathrm{K}_{fIP}^j \leq \mathrm{K}^{**j}_{fIP}$ and $\mathrm{M}_{gRP}^j \geq \mathrm{M}^{**j}_{gRP}$, then

$$CCIRIFDWG_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) \leq CCIRIFDWG_t(\mathfrak{H}^{**1}, \mathfrak{H}^{**2}, \ldots, \mathfrak{H}^{**Z})$$

$$CCIRIFDWG_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) \leq CCIRIFDWG_{tc}(\mathfrak{H}^{**1}, \mathfrak{H}^{**2}, \ldots, \mathfrak{H}^{**Z})$$

Property 6. (Boundedness) If $\mathfrak{H}^- = \begin{pmatrix} (min\mathrm{K}_{RP}^j, min\mathrm{K}_{IP}^j), (max\mathrm{M}_{RP}^j, max\mathrm{M}_{IP}^j), \\ \left(min\check{\Upsilon}_j^{RP}, min\check{\Upsilon}_j^{IP}\right) \end{pmatrix}$ and $\mathfrak{H}^+ = \begin{pmatrix} (max\mathrm{K}_{RP}^j, max\mathrm{K}_{IP}^j), (min\mathrm{M}_{RP}^j, min\mathrm{M}_{IP}^j), \\ \left(max\check{\Upsilon}_j^{RP}, max\check{\Upsilon}_j^{IP}\right) \end{pmatrix}, j = 1, 2, \ldots, Z$ thus.

$$\mathfrak{H}^- \leq CCIRIFDWG_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) \leq \mathfrak{H}^+$$

$$\mathfrak{H}^- \leq CCIRIFDWG_{tc}(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z) \leq \mathfrak{H}^+$$

The proof for t-conorm is same as for all proposed operators.

ARTICLE IN PRESS

ARTICLE IN PRESS

4. Proposed MCDM Framework

This section recapped the information on the formulation of the MCDM through the provision of the operators CCIRIFDWA and the CCIRIFDWG. Consider the alternatives $\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z$ with the attributes $\mathfrak{H}^1_{BT}, \mathfrak{H}^2_{BT}, \ldots, \mathfrak{H}^m_{BT}$ and the $\propto_j \in [0, 1]$ be the WV for each attribute with the condition $\sum_{j=1}^{Z} \propto_j = 1$. We construct a matrix that contains the data in the shape CCIRIFNs, the terms $\mathrm{K}_f(x) = \left(\mathrm{K}^j_{fRP}(x), \mathrm{K}^j_{fIP}(x)\right)$ be the degree of belonging and $\mathrm{M}_g(x) = \left(\mathrm{M}^j_{gRP}(x), \mathrm{M}^j_{gIP}(x)\right)$ is the degree of non-belonging, with the condition $0 \le \mathrm{K}^j_{fRP} + \mathrm{M}^j_{gRP}(x) \le 1$, $0 \le \mathrm{K}^j_{fIP} + \mathrm{M}^j_{gIP}(x) \le 1$. $\check{\mathrm{Y}}_o = \left(\check{\mathrm{Y}}^{PR}_o, \check{\mathrm{Y}}^{IP}_o\right) \in [0, 1] \times [0, 1]$ Represents the radial component around the circular domain to each element of $x \in \beta_\varepsilon$. Furthermore, the hesitancy degree is shaped as $\mathcal{H}(x) = \left(1 - \mathrm{K}^{RP}_f - \mathrm{M}^{RP}_g, 1 - \mathrm{K}^{IP}_f - \mathrm{M}^{IP}_g\right)$ a CCIRIFNs $\mathfrak{H}^j = \left(\left(\mathrm{K}^j_{fPR}, \mathrm{K}^j_{fIR}\right), \left(\mathrm{M}^j_{gPR}, \mathrm{M}^j_{gIR}\right), \left(\check{\mathrm{Y}}^{PR}_j, \check{\mathrm{Y}}^{IR}_j\right)\right), j = 1, 2, \ldots, Z$. Therefore, we begin the process of the MADM technique, the useful and productive steps of which are mentioned below:

Step 1: Organize the CCIRIFNs in a matrix. Additionally, we may have two types of information, such as cost and benefit. If we have the cost information, then normalize it accordingly.

$$Y = \begin{cases} \left(\left(\mathrm{K}^j_{fPR}, \mathrm{K}^j_{fIR}\right), \left(\mathrm{M}^j_{gPR}, \mathrm{M}^j_{gIR}\right), \left(\check{\mathrm{Y}}^{PR}_j, \check{\mathrm{Y}}^{IR}_j\right)\right) \; for\ benifit \\ \left(\left(\mathrm{K}^j_{fPR}, \mathrm{K}^j_{fIR}\right), \left(\mathrm{M}^j_{gPR}, \mathrm{M}^j_{gIR}\right), \left(\check{\mathrm{Y}}^{PR}_j, \check{\mathrm{Y}}^{IR}_j\right)\right) \; for\ cost \end{cases}$$

In the other one, the normalization of the matrix is not done

Step 2. The data can be aggregated using the CCIRIFDWA or CCIRIFDWG operator to obtain the aggregated information, as illustrated in the following example.

$CCIRIFDWA_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z)$

$$= \left\langle \left(\begin{array}{c} 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\mathrm{K}^1_{jRP}}{1 - \mathrm{K}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\mathrm{K}^1_{jIP}}{1 - \mathrm{K}^1_{jiP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right), \left(\begin{array}{c} \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \mathrm{M}^1_{gRP}}{\mathrm{M}^1_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \mathrm{M}^1_{gIP}}{\mathrm{M}^1_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right) \right\rangle$$

$$= \left\langle \left(\begin{array}{c} 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\check{\mathrm{Y}}^1_{jRP}}{1 - \check{\mathrm{Y}}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\check{\mathrm{Y}}^1_{jiP}}{1 - \check{\mathrm{Y}}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right) \right\rangle$$

$CCIRIFDWG_t(\mathfrak{H}^1, \mathfrak{H}^2, \ldots, \mathfrak{H}^Z)$

$$= \left\langle \left(\begin{array}{c} \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \mathrm{K}^1_{jRP}}{\mathrm{K}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \mathrm{K}^1_{jIP}}{\mathrm{K}^1_{jiP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right), \left(\begin{array}{c} 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\mathrm{M}^1_{gRP}}{1 - \mathrm{M}^1_{gRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ 1 - \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{\mathrm{M}^1_{gIP}}{1 - \mathrm{M}^1_{gIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right) \right\rangle$$

$$= \left\langle \left(\begin{array}{c} \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \check{\mathrm{Y}}^1_{jRP}}{\check{\mathrm{Y}}^1_{jRP}}\right)^{\tau}\right)^{\frac{1}{\tau}}}, \\ \dfrac{1}{1 + \left(\sum_{j=1}^{Z} \propto_j \left(\dfrac{1 - \check{\mathrm{Y}}^1_{jiP}}{\check{\mathrm{Y}}^1_{jIP}}\right)^{\tau}\right)^{\frac{1}{\tau}}} \end{array} \right) \right\rangle$$

Step 3: Compute the scores to reflect the above information, including

ARTICLE IN PRESS

$$S(\mathfrak{H}^j) = \left(\frac{\breve{\Upsilon}_{jRP} + \breve{\Upsilon}_{jIP}}{2}\right) * \left(\frac{\mathrm{K}_{jRP} - \mathrm{M}_{jRP} + \mathrm{K}_{jIP} - \mathrm{M}_{jIP}}{2}\right) \in [0, 1]$$

$$H(\mathfrak{H}^j) = \left(\frac{\breve{\Upsilon}_{jRP} + \breve{\Upsilon}_{jIP}}{2}\right) * \left(\frac{\mathrm{K}_{jRP} + \mathrm{M}_{jRP} + \mathrm{K}_{jIP} + \mathrm{M}_{jIP}}{2}\right) \in [0, 1]$$

Step 4: Final Ranking, to examine the superior decision, we rank the data according to their scores.

5. Case Study

Sustainable agriculture involves multidimensional decision-making amid uncertainty, cyclical environmental transformations, and competing goals, such as food security, climate adaptation, and socio-economic inclusion. Agricultural policymakers and rural development planners face the dilemma of evaluating and comparing different sustainability strategies and prioritizing them to make use of scarce resources. In this case study, we simulate a multi-criteria decision problem faced by a local agricultural planning unit in a climate-sensitive region, where the formally required task is to recommend ways to select the best intervention to improve productivity and resilience among smallholder farming groups. The main goal of the proposed MCDM model is to prioritize five potential agricultural strategies based on various sustainability criteria.

- $\mathfrak{H}^1$ Precision irrigation systems
- $\mathfrak{H}^2$ Organic compost-based nutrient management
- $\mathfrak{H}^3$ Crop rotation with resilient cultivars
- $\mathfrak{H}^4$ Controlled environment greenhouse farming
- $\mathfrak{H}^5$ Integrated pest and soil fertility management

These alternatives were evaluated across four critical sustainability attributes commonly observed in agricultural development assessments:

- $\mathfrak{H}_{BT}^1$ Resource efficiency
- $\mathfrak{H}_{BT}^2$ Environmental impact mitigation
- $\mathfrak{H}_{BT}^3$ Economic feasibility
- $\mathfrak{H}_{BT}^4$ Adaptability to climate variability
- $\mathfrak{H}_{BT}^5$ Social acceptability among smallholder farmers

Using the given alternatives, we aim to determine the optimal solution, which requires the various attributes provided above. Furthermore, we assign the weight vectors (WV) to attributes in order $(0.1, 0.2, 0.1, 0.2, 0.4)^T$. The MADM methodology is initiated as follows.

Step 1. Table 2 is based on expert assumptions regarding CCIRIFNs. All criteria are of the benefit type, so no normalization is needed. A hypothetical expert-based dataset is used in this study to recreate a realistic scenario of agricultural decision-making, allowing the model to be evaluated. Values are presented as CCIRIFNs.

Table 2. CCIRIF Decision Matrix

Alternatives	$\mathfrak{H}_{BT}^1$	$\mathfrak{H}_{BT}^2$	$\mathfrak{H}_{BT}^3$	$\mathfrak{H}_{BT}^4$	$\mathfrak{H}_{BT}^5$
$\mathfrak{H}^1$	$\begin{pmatrix}(0.6, 0.3)\\(0.2, 0.2)\\(0.2, 0.1)\end{pmatrix}$	$\begin{pmatrix}(0.61, 0.31)\\(0.21, 0.21)\\(0.21, 0.11)\end{pmatrix}$	$\begin{pmatrix}(0.62, 0.32)\\(0.22, 0.22)\\(0.22, 0.12)\end{pmatrix}$	$\begin{pmatrix}(0.63, 0.33)\\(0.23, 0.23)\\(0.23, 0.13)\end{pmatrix}$	$\begin{pmatrix}(0.64, 0.34)\\(0.24, 0.24)\\(0.24, 0.14)\end{pmatrix}$
$\mathfrak{H}^2$	$\begin{pmatrix}(0.5, 0.1)\\(0.3, 0.2)\\(0.2, 0.1)\end{pmatrix}$	$\begin{pmatrix}(0.51, 0.11)\\(0.31, 0.21)\\(0.21, 0.11)\end{pmatrix}$	$\begin{pmatrix}(0.52, 0.12)\\(0.32, 0.22)\\(0.22, 0.12)\end{pmatrix}$	$\begin{pmatrix}(0.53, 0.13)\\(0.33, 0.23)\\(0.23, 0.13)\end{pmatrix}$	$\begin{pmatrix}(0.54, 0.14)\\(0.34, 0.24)\\(0.24, 0.14)\end{pmatrix}$
$\mathfrak{H}^3$	$\begin{pmatrix}(0.5, 0.4)\\(0.3, 0.3)\\(0.1, 0.2)\end{pmatrix}$	$\begin{pmatrix}(0.51, 0.41)\\(0.31, 0.31)\\(0.11, 0.21)\end{pmatrix}$	$\begin{pmatrix}(0.52, 0.42)\\(0.32, 0.32)\\(0.12, 0.22)\end{pmatrix}$	$\begin{pmatrix}(0.53, 0.43)\\(0.33, 0.33)\\(0.13, 0.23)\end{pmatrix}$	$\begin{pmatrix}(0.54, 0.44)\\(0.34, 0.34)\\(0.14, 0.24)\end{pmatrix}$
$\mathfrak{H}^4$	$\begin{pmatrix}(0.6, 0.4)\\(0.3, 0.3)\\(0.3, 0.1)\end{pmatrix}$	$\begin{pmatrix}(0.61, 0.41)\\(0.31, 0.31)\\(0.31, 0.11)\end{pmatrix}$	$\begin{pmatrix}(0.62, 0.42)\\(0.32, 0.32)\\(0.32, 0.12)\end{pmatrix}$	$\begin{pmatrix}(0.63, 0.43)\\(0.33, 0.33)\\(0.33, 0.13)\end{pmatrix}$	$\begin{pmatrix}(0.64, 0.44)\\(0.34, 0.34)\\(0.34, 0.14)\end{pmatrix}$
$\mathfrak{H}^5$	$\begin{pmatrix}(0.7, 0.5)\\(0.1, 0.3)\\(0.2, 0.1)\end{pmatrix}$	$\begin{pmatrix}(0.71, 0.51)\\(0.11, 0.31)\\(0.21, 0.11)\end{pmatrix}$	$\begin{pmatrix}(0.72, 0.52)\\(0.12, 0.32)\\(0.22, 0.12)\end{pmatrix}$	$\begin{pmatrix}(0.73, 0.53)\\(0.13, 0.33)\\(0.23, 0.13)\end{pmatrix}$	$\begin{pmatrix}(0.74, 0.54)\\(0.14, 0.34)\\(0.24, 0.14)\end{pmatrix}$

ARTICLE IN PRESS

Step 2: We aggregate the data to proceed with the CCIRIFDWA and CCIRIFDWG operators, as shown in Table 3.

Table 3. Aggregated values for proposed operators

Alternatives	CCIRIFDWA	CCIRIFDWG
$\mathfrak{H}^1$	$\begin{pmatrix}(0.6269, 0.3267)\\(0.2244, 0.2244)\\(0.2268, 0.1271)\end{pmatrix}$	$\begin{pmatrix}(0.6252, 0.3248)\\(0.2268, 0.2268)\\(0.2244, 0.1232)\end{pmatrix}$
$\mathfrak{H}^2$	$\begin{pmatrix}(0.5262, 0.1271)\\(0.3248, 0.2244)\\(0.2268, 0.1271)\end{pmatrix}$	$\begin{pmatrix}(0.5251, 0.1232)\\(0.3267, 0.2268)\\(0.2244, 0.1232)\end{pmatrix}$
$\mathfrak{H}^3$	$\begin{pmatrix}(0.5268, 0.4267)\\(0.3248, 0.3248)\\(0.1271, 0.2268)\end{pmatrix}$	$\begin{pmatrix}(0.5251, 0.4250)\\(0.3267, 0.3267)\\(0.1232\ 0.2244)\end{pmatrix}$
$\mathfrak{H}^4$	$\begin{pmatrix}(0.6269, 0.4268)\\(0.3248, 0.3248)\\(0.3276, 0.1271)\end{pmatrix}$	$\begin{pmatrix}(0.6252, 0.4250)\\(0.3267, 0.3267)\\(0.1232\ 0.2244)\end{pmatrix}$
$\mathfrak{H}^5$	$\begin{pmatrix}(0.6269, 0.3267)\\(0.2244, 0.2244)\\(0.2268, 0.1271)\end{pmatrix}$	$\begin{pmatrix}(0.6252, 0.3248)\\(0.1271, 0.3267)\\(0.2244, 0.1232)\end{pmatrix}$

Step 3. Compute the score values for the numbers obtained, as shown in Table 4.

Table 4. Score values for proposed operators

Alternatives	CCIRIFDWA	CCIRIFDWG
$\mathfrak{H}^1$	(0.046)	(0.042)
$\mathfrak{H}^2$	(0.009)	(0.008)
$\mathfrak{H}^3$	(0.025)	(0.028)
$\mathfrak{H}^4$	(0.046)	(0.045)
$\mathfrak{H}^5$	(0.072)	(0.069)

Step 4. Lastly, to follow the best choice, we rank the data according to their scores as presented in Table 5.

ARTICLE IN PRESS

Table 5. Score values for proposed operators

Framework	Ranking	Best one
CCIRIFDWA	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
CCIRIFDWG	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$

$\mathfrak{H}^5$ ranked highest according to the aggregated CCIRIF scores had the best scores in adaptability and social acceptance to the set weightings. The indices based on the two operators (CCIRIFDWA and CCIRIFDWG) provide important insights into the sustainability priorities of agricultural systems. $\mathfrak{H}^5$ The most desired strategy turns out to be more efficient than the others, as it is balanced across economic, environmental, and resource-based parameters, indicating that it can enhance productivity while alleviating ecological pressures. The second one is a strategy $\mathfrak{H}^4$, which is mainly due to its flexibility and resource efficiency. On the other hand, $\mathfrak{H}^2$ is the least because the level of performance has been relatively poor in the parameters related to environmental resilience and long-term sustainability, which are normally of paramount importance in real-world agricultural decision-making. The outcomes of the above-presented results are consistent with the sustainability-related

ARTICLE IN PRESS

problems pursued by the farming sector in modern times, i.e., the need to balance resource efficiency, adapt to climate change, and make economically viable decisions, which proves the fact that the proposed model reflects the multidimensional nature of the agricultural sector of interest. The proposed CCIRIFD operator is a scalable, computationally efficient model that can be used in larger or more complex decision-making contexts. Aggregation is based on simple arithmetic and exponential functions, and computationally it is not complicated (computational complexity). $O(nxm)$, n is the number of choices and m is the number of criteria. This linear scalability enables the model to process larger datasets effectively without significant increases in processing time. In addition, the modularity of the Dombi operators can be easily extended to multi-stage or hierarchical decision-making systems, in which intermediate evaluations may be aggregated at the next level. This flexibility makes the method applicable in real-world applications, such as the planning of regions' agricultural resources or environmental sustainability analysis, which involve many layers of criteria and expert judgment. See figure 2

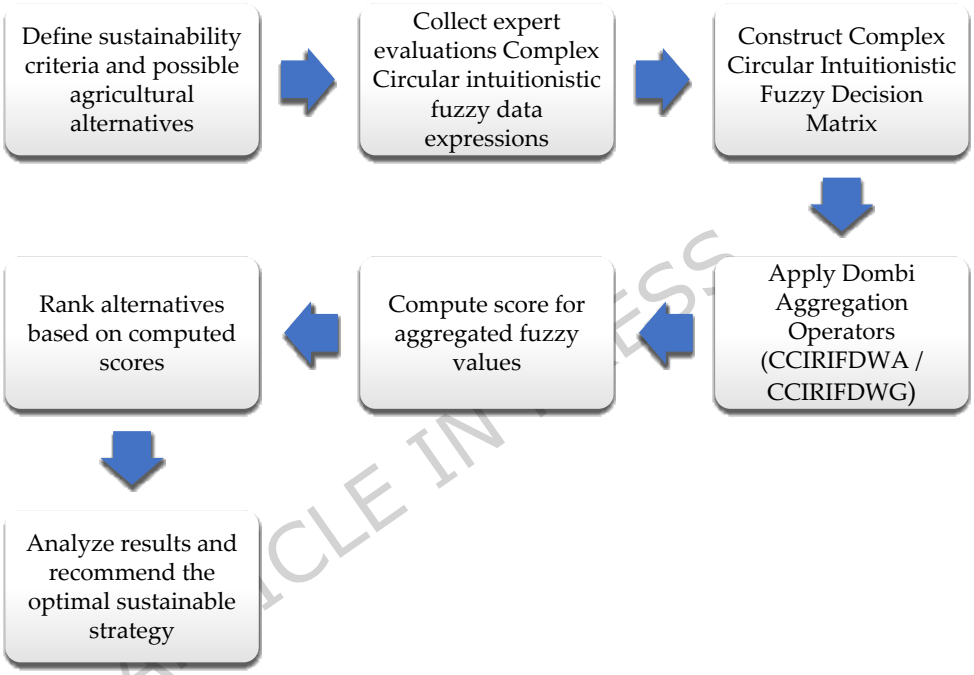


Figure 2. Step-wise guide for the proposed methodology

6. Comparative Study

To compare the performance and stability of the proposed algorithm with the existing literature across various environments, several algorithms from the literature have been employed, and their performance has been analyzed using relevant data. Seikh and Mandal [56] and Liu et al. [57] applied the Dombi operator in the context of IFSs and Dombi Bonferroni mean operators. Li et al. [58] employed Dombi-Hamy mean operators, and Wang et al. [59] utilized Dombi prioritized operators, both of which exhibit IFS characteristics. The results related to them are presented in Table 5. Using this Table, it is evident that the most preferred option identified under every method is $\mathfrak{H}^5$. This indicates that the given approach is conservative. At the same time, it has also been observed through this Table that the current methods cannot differentiate among the alternatives $\mathfrak{H}^2$ and $\mathfrak{H}^3$. To put it another way, once these existing methods are applied, one will be able to discover that a difference does not exist between $\mathfrak{H}^2$ and $\mathfrak{H}^3$. Therefore, the existing methods cannot properly prioritize the alternatives. Moreover, the technique suggested under the operator variation also provides an acceptable ranking among the alternatives. This discussion implies that it is imperative to assume ratings for CCIRIFNs over other existing environments. Table 6 presents the comparative analysis. Table 7 presents a theoretical comparison of current approaches and the proposed CCIRIF-based operators. It emphasizes that the

ARTICLE IN PRESS

proposed models are designed to accommodate radius functions and two-dimensional information, making them more global and effective in complex decision-making situations.

Table 6. Comparative analysis with existing studies

Framework	Ranking	Best one
CCIRIFDWA	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
CCIRIFDWG	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	
Seikh and Mandal [56]	$\mathfrak{H}^5 > \mathfrak{H}^1 > \mathfrak{H}^4 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
Liu et al [57]	$\mathfrak{H}^5 > \mathfrak{H}^1 > \mathfrak{H}^4 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
Li et al. [58]	$\mathfrak{H}^5 > \mathfrak{H}^1 > \mathfrak{H}^4 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
Wang et al. [59]	$\mathfrak{H}^5 > \mathfrak{H}^1 > \mathfrak{H}^4 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$

Table 7. Theoretical Comparison of the Proposed Work

Methods	Truth function	Falsity function	Radius function	2-D information	AOs	Dombi operators
Proposed operators	✓	✓	✓	✓	✓	✓
Seikh and Mandal **[56]**	✓	✓	×	×	✓	✓
Liu et al **[57]**	✓	✓	×	×	✓	✓
Li et al. **[58]**	✓	✓	×	×	✓	✓
Wang et al. **[59]**	✓	✓	×	×	✓	✓

6.1. Sensitivity Analysis

In this section, the sensitivity of the weight variation to the aggregate operators of the postulated decision-making framework was examined to confirm the soundness and stability of both the CCIRIFDWA and the CCIRIFDWG.

The weight basis of the base weight calculated in this investigation was: $(0.1, 0.2, 0.1, 0.2, 0.4)^T$, which are the five analysis criteria considered. The weights were varied systematically over a small range, with the normalized sum of the weights held constant. This method will enable the stability of the ranking results to be evaluated under moderate changes in the relative significance of the decision criteria. The values obtained through aggregation were recalculated for each variation, and the resulting ranking of the alternatives was compared with the initial ranking. The analysis also ensures that the model's decision-making outcomes are not highly sensitive to variations in the given criterion weights, which justifies its strength in meeting the requirements of uncertainty conditions. See Table 8 and Figure 3.

Table 8. Sensitivity Analysis of Operator CCIRIFDWA

Adjusted Weights (Normalized)	Ranking of Alternatives	Optimal
$[\mathbf{0.10}, \mathbf{0.20}, \mathbf{0.10}, \mathbf{0.20}, \mathbf{0.40}]$	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
$[\mathbf{0.11}, \mathbf{0.18}, \mathbf{0.09}, \mathbf{0.22}, \mathbf{0.40}]$	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
$[\mathbf{0.09}, \mathbf{0.22}, \mathbf{0.11}, \mathbf{0.18}, \mathbf{0.40}]$	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
$[\mathbf{0.12}, \mathbf{0.16}, \mathbf{0.08}, \mathbf{0.24}, \mathbf{0.40}]$	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
$[\mathbf{0.08}, \mathbf{0.24}, \mathbf{0.12}, \mathbf{0.16}, \mathbf{0.40}]$	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$

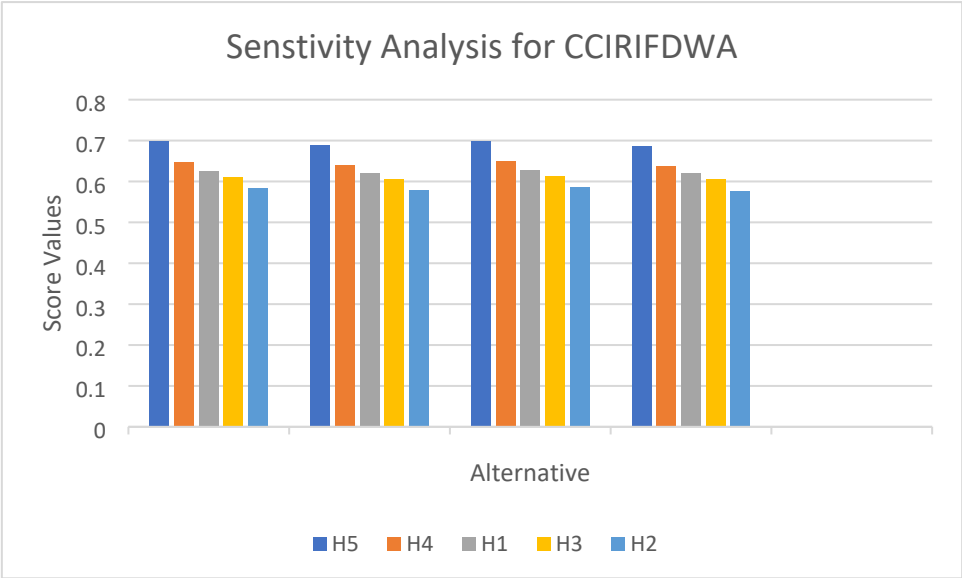
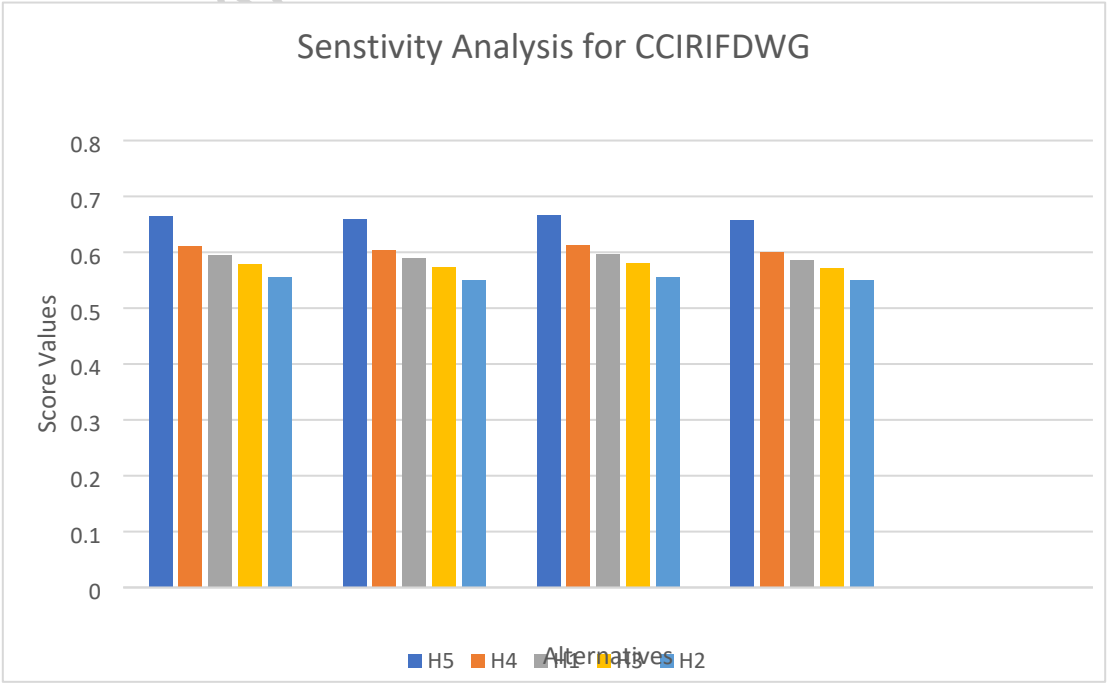


Figure 3. Sensitivity analysis for the CCIRIFDWA operator

The ranking sequence obtained with the CCIRIFDWA operator was also entirely consistent across all weight variations. Although the relative significance of the assessment criteria changed moderately, the alternative with the highest performance $\mathfrak{H}^5$ was always the first, and then came $\mathfrak{H}^4$ and $\mathfrak{H}^1$. This is the homogeneity in rank, indicating that the CCIRIFDWA operator provides a secure and reliable framework for aggregation, which may yield consistent, clearly made decisions even when input weights differ within reasonable bounds.

Table 9. Sensitivity Analysis of Operator CCIRIFDWA

Adjusted Weights (Normalized)	**Ranking of Alternatives**	Optimal
[0.10, 0.20, 0.10, 0.20, 0.40]	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
[0.11, 0.18, 0.09, 0.22, 0.40]	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
[0.09, 0.22, 0.11, 0.18, 0.40]	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
[0.12, 0.16, 0.08, 0.24, 0.40]	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$
[0.08, 0.24, 0.12, 0.16, 0.40]	$\mathfrak{H}^5 > \mathfrak{H}^4 > \mathfrak{H}^1 > \mathfrak{H}^3 > \mathfrak{H}^2$	$\mathfrak{H}^5$



ARTICLE IN PRESS

Figure 4. Sensitivity analysis for the CCIRIFDWG operator

In Table 9 and Figure 4, the CCIRIFDWG operator analysis yielded the same results. In all the weight adjustment conditions, the optimum alternative $\mathfrak{H}^5$ did not change, indicating that the model was not sensitive to parameter variations. There are slight variations in outcomes that do not alter the overall preference order, suggesting that geometric aggregation with Dombi effectively preserves ranking integrity when the weights are variable.

The comparative sensitivity analysis of the two operators shows that the provided CCIRIFDWA and CCIRIFDWG frameworks are robust and resilient. The consistency of the rankings across various weighting models indicates that the decision-making process is not disproportionately affected by small fluctuations in the importance of the criterion. The model will therefore provide decision-makers with high confidence when applied to actual sustainability appraisal in the real world, where uncertainty in expert judgements and weighting preferences cannot be eliminated.

Note that the results of the proposed model are obtained using MS Excel. However, MATLAB and Python can also be used for this purpose. In addition, the model's time complexity depends on the number of alternatives and attributes. However, once the code is run once, it can be used for a larger number of alternatives and attributes.

7. Conclusions

This paper presented a CCIRIF multi-criteria decision-making model to assess sustainable agricultural policies in the presence of uncertainties. CCIRIFDWA and Geometric CCIRIFDWG operators: Two new aggregation operators were created to handle two-dimensional and cyclic uncertainties better. The suggested framework was implemented in a theoretical yet knowledge-based case study of five sustainability criteria and five agricultural strategies. The results identified controlled-environment greenhouse farming as the best alternative, indicating that the model is very effective at capturing ambiguity and interdependence in the complex decision environment.

The research adds a versatile, mathematically consistent, and interpretable decision-support method to improve sustainability assessment in agriculture and other areas. It should be noted, however, that the present evaluation is based on hypothetical data and fixed-weight assumptions, which might limit generalizability. Future studies will focus on establishing the model using real-world data, applying adaptive weighting techniques, and expanding the model to dynamic, multi-stage decision-making. Such trends will only make the model more applicable and effective in achieving sustainable planning and policy formulation to support agriculture.

ARTICLE IN PRESS

7.1 Limitations and Future Work

Although the introduced framework has immense capacity to model uncertainty and to capture cyclic models to facilitate long-term, sustainable farming decisions, there are a number of limitations that must be considered. The study is grounded in hypothetical and expert knowledge. It assumes that the assessment criteria's weight values are fixed, which might not fully reflect the variability dynamics of agricultural systems. Additionally, the current model has not been well established by empirical research. Future research will thus involve implementing the framework on real-world agricultural variables, determining weights by incorporating stakeholders, and generalising the framework to dynamic, multi-period, and data-driven decision environments. This paper assumes the data are hypothetical, with values based on professional estimates, and may not accurately reflect actual variability in agriculture. The one-stage decision stage, coupled with fixed weight assignments, also facilitates generalizability. Future possibilities include developing the model on empirical grounds and in dynamic, multi-stage decision environments. A further improvement of the framework that would make it more relevant and applicable would be the integration of stakeholder feedback and some real-time agricultural data. Though the current study uses hypothetical data to confirm the conceptual soundness and computational efficiency of the suggested model, future studies can empirically support the framework by incorporating real-world data or expert-derived measurements. Such information

ARTICLE IN PRESS

would be included, enabling greater control over the parameters, enhancing the interpretability of the results, and providing comparative verification of the model against observed performance indicators. The proposed framework is adaptable and can easily accommodate empirical data from an agricultural or environmental system, enhancing its applicability and practical utility in future studies. The suggested framework has great potential to guide sustainability policies and feasible farm-level planning by allowing prioritization of strategies that could enhance climate resilience, more efficient resource use, and long-term sustainability in agricultural planning, using data.

Author's contribution. Yanfeng Yu contributed fully to the completion of the manuscript.

Funding. This work was supported by the National Key Research and Development Sub-Project of China (No.2022YFD1600604-03), the Fundamental Research and Talent Training Project of Jiangxi Academy of Agricultural Sciences, China (No.JXSNKYJCRC202317), Jiangxi Province Modern Agriculture (Rice & Rape) Industrial Technology System Special Project, China (No.JXARS-O4&JXARS-08).

Data Availability Statement: The data are unavailable due to privacy or ethical restrictions.

Conflicts of interest. The author has no conflicts of interest.

Acknowledgment: AI tools were used exclusively for grammar correction and text polishing. The authors reviewed and approved all generated content and take full responsibility for the final manuscript.

References

[1] M. Yazdani, E. D. Gonzalez, and P. Chatterjee, "A multi-criteria decision-making framework for agriculture supply chain risk management under a circular economy context," *Management Decision*, vol. 59, no. 8, pp. 1801–1826, 2021.

[2] H. Li, Y. Zhao, and F. Zheng, "The framework of an agricultural land-use decision support system based on ecological environmental constraints," *Science of The Total Environment*, vol. 717, p. 137149, May 2020, doi: 10.1016/j.scitotenv.2020.137149.

[3] B. D. Rouyendegh and Ş. Savalan, "An Integrated Fuzzy MCDM Hybrid Methodology to Analyze Agricultural Production," *Sustainability*, vol. 14, no. 8, Art. no. 8, Jan. 2022, doi: 10.3390/su14084835.

[4] L. Thi, C.-N. Wang, and N. Thanh, "Fuzzy MCDM for Improving the Performance of Agricultural Supply Chain," *CMC*, vol. 73, no. 2, pp. 4003–4015, 2022, doi: 10.32604/cmc.2022.030209.

[5] G. Demir, "Selection of Biomass Resources with Fuzzy Multi-Criteria Decision-Making Methods: Alternatives for Sustainable Energy Production," *Journal of Operations Intelligence*, vol. 3, no. 1, Art. no. 1, Apr. 2025, doi: 10.31181/jopi31202537.

[6] L. Balezentiene, D. Streimikiene, and T. Balezentis, "Fuzzy decision support methodology for sustainable energy crop selection," *Renewable and Sustainable Energy Reviews*, vol. 17, pp. 83–93, Jan. 2013, doi: 10.1016/j.rser.2012.09.016.

[7] J. Więckowski, J. Wątróbski, and W. Sałabun, "Toward robust decision-making under multiple evaluation scenarios with a novel fuzzy ranking approach: green supplier selection study case," *Artif Intell Rev*, vol. 58, no. 1, p. 3, Nov. 2024, doi: 10.1007/s10462-024-11006-8.

[8] C. Kahraman, S. C. Onar, and B. Oztaysi, "Fuzzy Multicriteria Decision-Making: A Literature Review:," *IJCIS*, vol. 8, no. 4, p. 637, 2015, doi: 10.1080/18756891.2015.1046325.

[9] A. Hussain, T. Mahmood, F. Smarandache, and S. Ashraf, "TOPSIS approach for MCGDM based on intuitionistic fuzzy rough Dombi aggregation operations," *Computational and Applied Mathematics*, vol. 42, no. 4, p. 176, 2023.

[10] R. J. Girubha and S. Vinodh, "Application of fuzzy VIKOR and environmental impact analysis for material selection of an automotive component," *Materials & Design*, vol. 37, pp. 478–486, 2012.

[11] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338–353, June 1965, doi: 10.1016/S0019-9958(65)90241-X.

ARTICLE IN PRESS

[12] K. T. Atanassov, "Intuitionistic fuzzy sets," *Fuzzy Sets and Systems*, vol. 20, no. 1, pp. 87–96, Aug. 1986, doi: 10.1016/S0165-0114(86)80034-3.

[13] B. A. Hashsmi, "Multi-Criteria Decision-Making Approach with Interval-Valued Intuitionistic Fuzzy Assessment for Green Supplier Evaluation and Selection," thesis, Toronto Metropolitan University, 2023. doi: 10.32920/24058713.v1.

[14] M. R. Seikh and U. Mandal, "Intuitionistic fuzzy Dombi aggregation operators and their application to multiple attribute decision-making," *Granul. Comput.*, vol. 6, no. 3, pp. 473–488, July 2021, doi: 10.1007/s41066-019-00209-y.

[15] M. R. Khan, K. Ullah, Q. Khan, and D. Pamucar, "Intuitionistic fuzzy dombi aggregation information involving lower and upper approximations," *Comp. Appl. Math.*, vol. 44, no. 3, p. 144, Jan. 2025, doi: 10.1007/s40314-024-03044-3.

[16] X. Kong, "Complex Circular Intuitionistic Fuzzy Heronian Mean Aggregation for Dynamic Air Quality Monitoring and Public Health Risk Prediction," *IEEE Access*, vol. 12, pp. 129897–129916, 2024, doi: 10.1109/ACCESS.2024.3458172.

[17] R. R. Yager, "Pythagorean fuzzy subsets," in *2013 joint IFSA world congress and NAFIPS annual meeting (IFSA/NAFIPS)*, IEEE, 2013, pp. 57–61.

[18] R. R. Yager, "Generalized Orthopair Fuzzy Sets," *IEEE Transactions on Fuzzy Systems*, vol. 25, no. 5, pp. 1222–1230, Oct. 2017, doi: 10.1109/TFUZZ.2016.2604005.

[19] N. El-Hefenawy, M. A. Metwally, Z. M. Ahmed, and I. M. El-Henawy, "A Review on the Applications of Neutrosophic Sets," *Journal of Computational and Theoretical Nanoscience*, vol. 13, no. 1, pp. 936–944, Jan. 2016, doi: 10.1166/jctn.2016.4896.

[20] J. Ye, "A multicriteria decision-making method using aggregation operators for simplified neutrosophic sets," *Journal of Intelligent & Fuzzy Systems*, vol. 26, no. 5, pp. 2459–2466, 2014.

[21] G. Beliakov, S. James, J. Mordelová, T. Rückschlossová, and R. R. Yager, "Generalized Bonferroni mean operators in multi-criteria aggregation," *Fuzzy Sets and Systems*, vol. 161, no. 17, pp. 2227–2242, Sept. 2010, doi: 10.1016/j.fss.2010.04.004.

[22] W. Azeem, W. Mahmood, T. Mahmood, and Z. Ali, "Analysis of Einstein aggregation operators based on complex intuitionistic fuzzy sets and their applications in multi-attribute decision-making," *MATH*, vol. 8, no. 3, Art. no. math-08-03-305, 2023, doi: 10.3934/math.2023305.

[23] S. Abdullah, M. Qiyas, M. Naeem, and Y. Liu, "Pythagorean Cubic fuzzy Hamacher aggregation operators and their application in green supply selection problem," *AIMS Mathematics*, vol. 7, no. 3, pp. 4735–4766, 2022.

[24] Z. Ali, W. Emam, T. Mahmood, and H. Wang, "Archimedean Heronian mean operators based on complex intuitionistic fuzzy sets and their applications in decision-making problems," *Heliyon*, vol. 10, no. 3, 2024.

[25] C. Xu, "New measure of circular intuitionistic fuzzy sets and its application in decision making," *AIMS Mathematics*, vol. 8, no. 10.

[26] H. Garg and D. Rani, "Generalized Geometric Aggregation Operators Based on T-Norm Operations for Complex Intuitionistic Fuzzy Sets and Their Application to Decision-making," *Cogn Comput*, vol. 12, no. 3, pp. 679–698, May 2020, doi: 10.1007/s12559-019-09678-4.

[27] K. T. Atanassov, "Circular intuitionistic fuzzy sets," *Journal of Intelligent & Fuzzy Systems*, vol. 39, no. 5, pp. 5981–5986, Nov. 2020, doi: 10.3233/JIFS-189072.

[28] M. R. N. Qureshi, R. K. Singh, and Mohd. A. Hasan, "Decision support model to select crop pattern for sustainable agricultural practices using fuzzy MCDM," *Environ Dev Sustain*, vol. 20, no. 2, pp. 641–659, Apr. 2018, doi: 10.1007/s10668-016-9903-7.

[29] S. S. Evangelista *et al.*, "An Integrated Fermatean Fuzzy Multi-attribute Evaluation of Digital Technologies for Circular Public Sector Supply Chains," *Int J Comput Intell Syst*, vol. 16, no. 1, p. 122, July 2023, doi: 10.1007/s44196-023-00294-7.

[30] S. K. Sahoo and S. S. Goswami, "A Comprehensive Review of Multiple Criteria Decision-Making (MCDM) Methods: Advancements, Applications, and Future Directions," *Decision Making Advances*, vol. 1, no. 1, Art. no. 1, June 2023, doi: 10.31181/dma1120237.

ARTICLE IN PRESS

ARTICLE IN PRESS

[31] M. Amin, K. Ullah, M. Akram, R. Imran, and M. S. Nazeer, "Advancing AI-Based Biometric Authentication in Multi-Criteria Decision Approach Using Complex Circular Intuitionistic Fuzzy Logic and Dombi Operators," June 10, 2024, *Research Square*. doi: 10.21203/rs.3.rs-4483111/v1.

[32] T. Mahmood and Z. Ali, "Dombi Hamy Mean Operators Based on Complex Intuitionistic Fuzzy Uncertainty and Their Application in Multi-Attribute Decision-Making," in *Fuzzy Optimization, Decision-making and Operations Research: Theory and Applications*, C. Jana, M. Pal, G. Muhiuddin, and P. Liu, Eds., Cham: Springer International Publishing, 2023, pp. 257–280. doi: 10.1007/978-3-031-35668-1_13.

[33] M. Riaz and H. M. A. Farid, "Enhancing Green Supply Chain Efficiency Through Linear Diophantine Fuzzy Soft-Max Aggregation Operators," Mar. 2023, doi: 10.56578/jii010102.

[34] A. Saha, J. Reddy, and R. Kumar, "A fuzzy similarity based classification with Archimedean-Dombi aggregation operator," *Journal of Intelligent Management Decision*, vol. 1, no. 2, pp. 118–27, 2022.

[35] M. Sarfarz and D. Božanić, "Using Sugeno-Weber triangular norm-based interval value spherical fuzzy information for recycled water," *Acadlore Trans. Appl Math. Stat*, vol. 2, no. 3, pp. 117–132, 2024.

[36] M. A. Khan and K. Rahman, "Mathematical model to evaluate of client behavior in the automobile sector based on T-spherical fuzzy information within Dombi environment," *Acadlore Trans. Appl Math. Stat*, vol. 3, no. 1, pp. 52–64, 2025.

[37] U. Ahmad, "Selective image segmentation through fuzzy einstein–dombi operators and level set energy minimization," *Int J. Knowl. Innov Stud*, vol. 3, no. 2, pp. 74–88, 2025.

[38] C. Jana and M. Pal, "Interval-valued picture fuzzy uncertain linguistic dombi operators and their application in industrial fund selection," *J. Ind. Intell*, vol. 1, no. 2, pp. 110–124, 2023.

[39] S. Khan, M. I. Syam, H. A. Abujabal, M. Rahim, A. Alburaikan, and H. A. E.-W. Khalifa, "A strategic decision-making framework for evaluating barriers to green supply chain management using fractional fuzzy similarity measures," *Scientific Reports*, vol. 15, no. 1, p. 29089, 2025.

[40] S. Khan, H. A. Abujabal, M. Rahim, A. Almutairi, A. Alburaikan, and H. A. E.-W. Khalifa, "Multi attribute group decision-making based on quasirung orthopair fuzzy Frank aggregation operators for optimal vehicle selection," *Scientific Reports*, vol. 15, no. 1, p. 28868, 2025.

[41] H. Garg and U. ur Rehman, "A Group Decision-making Algorithm to Analyses Risk Evaluation of Hepatitis with Sine Trigonometric Laws under Bipolar Complex Fuzzy Sets Information.," *Journal of Innovative Research in Mathematical and Computational Sciences*, vol. 3, no. 1, pp. 65–90, 2024.

[42] A. Hussain, C. A. Raza, and K. Ulllah, "Advanced Decision-Making With Complex q-Rung Orthopair Fuzzy Muirhead Mean Models," *Spectrum of Decision Making and Applications*, vol. 3, no. 1, pp. 316–338, 2026.

[43] U. ur Rehman, T. Mahmood, and H. M. Waqas, "Managing uncertainty in battery energy storage system evaluation using complex hesitant fuzzy multi-criteria decision-making technique with Einstein operators," *Journal of Energy Storage*, vol. 125, p. 116968, 2025.

[44] B. Kizielewicz, J. Wątróbski, and W. Sałabun, "Multi-criteria decision support system for the evaluation of UAV intelligent agricultural sensors," *Artif Intell Rev*, vol. 58, no. 7, p. 194, Apr. 2025, doi: 10.1007/s10462-025-11201-1.

[45] A. Razzaque, A. Razaq, A. Khalid, I. Masmali, U. Shuaib, and G. Alhamzi, "Selecting an Optimal Approach to Reduce Drivers of Climate Change in a Complex Intuitionistic Fuzzy Environment," *Sustainability*, vol. 15, no. 16, Art. no. 16, Jan. 2023, doi: 10.3390/su151612300.

[46] Z. Ciric P, D. Stojic, O. Sedlak, A. Marcikic Horvat, and Z. Kleut, "Innovation Model of Agricultural Technologies Based on Intuitionistic Fuzzy Sets," *Sustainability*, vol. 11, no. 19, Art. no. 19, Jan. 2019, doi: 10.3390/su11195457.

[47] H. Garg and D. Rani, "Some results on information measures for complex intuitionistic fuzzy sets," *International Journal of Intelligent Systems*, vol. 34, no. 10, pp. 2319–2363, 2019, doi: 10.1002/int.22127.

ARTICLE IN PRESS

ARTICLE IN PRESS

[48] S. Faizi, W. Sałabun, T. Rashid, S. Zafar, and J. Wątróbski, "Intuitionistic Fuzzy Sets in Multi-Criteria Group Decision Making Problems Using the Characteristic Objects Method," *Symmetry*, vol. 12, no. 9, Art. no. 9, Sept. 2020, doi: 10.3390/sym12091382.

[49] J. Streimikis, "Comparative assessment of circular economy performance in the Baltic States using MCDM methods," *Transformations and Sustainability*, vol. 1, no. 1, Art. no. 1, May 2025, doi: 10.63775/pcxj8p61.

[50] E. Çakır and M. A. Taş, "Circular intuitionistic fuzzy decision making and its application," *Expert Systems with Applications*, vol. 225, p. 120076, 2023.

[51] N. Alkan and C. Kahraman, "An intuitionistic fuzzy multi-distance based evaluation for aggregated dynamic decision analysis (IF-DEVADA): Its application to waste disposal location selection," *Engineering Applications of Artificial Intelligence*, vol. 111, p. 104809, May 2022, doi: 10.1016/j.engappai.2022.104809.

[52] E. Çakır, M. A. Taş, and Z. Ulukan, "Circular Intuitionistic Fuzzy Sets in Multi Criteria Decision Making," in *11th International Conference on Theory and Application of Soft Computing, Computing with Words and Perceptions and Artificial Intelligence - ICSCCW-2021*, vol. 362, R. A. Aliev, J. Kacprzyk, W. Pedrycz, M. Jamshidi, M. Babanli, and F. M. Sadikoglu, Eds., in Lecture Notes in Networks and Systems, vol. 362. , Cham: Springer International Publishing, 2022, pp. 34–42. doi: 10.1007/978-3-030-92127-9_9.

[53] D. Ramot, R. Milo, M. Friedman, and A. Kandel, "Complex fuzzy sets," *IEEE transactions on fuzzy systems*, vol. 10, no. 2, pp. 171–186, 2002.

[54] A. M. J. S. Alkouri and A. R. Salleh, "Complex intuitionistic fuzzy sets," in *AIP conference proceedings*, American Institute of Physics, 2012, pp. 464–470.

[55] J. Dombi, "A general class of fuzzy operators, the demorgan class of fuzzy operators and fuzziness measures induced by fuzzy operators," *Fuzzy Sets and Systems*, vol. 8, no. 2, pp. 149–163, Aug. 1982, doi: 10.1016/0165-0114(82)90005-7.

[56] M. R. Seikh and U. Mandal, "Intuitionistic fuzzy Dombi aggregation operators and their application to multiple attribute decision-making," *Granul. Comput.*, vol. 6, no. 3, pp. 473–488, July 2021, doi: 10.1007/s41066-019-00209-y.

[57] P. Liu, J. Liu, and S.-M. Chen, "Some intuitionistic fuzzy Dombi Bonferroni mean operators and their application to multi-attribute group decision making," *Journal of the Operational Research Society*, pp. 1–26, 2017.

[58] Z. Li, H. Gao, and G. Wei, "Methods for multiple attribute group decision making based on intuitionistic fuzzy Dombi Hamy mean operators," *Symmetry*, vol. 10, no. 11, p. 574, 2018.

[59] P. Wang, B. Zhu, Y. Yu, Z. Ali, and B. Almohsen, "COMPLEX INTUITIONISTIC FUZZY DOMBI PRIORITIZED AGGREGATION OPERATORS AND THEIR APPLICATION FOR RESILIENT GREEN SUPPLIER SELECTION," *Facta Universitatis, Series: Mechanical Engineering*, vol. 21, no. 3, Art. no. 3, Oct. 2023, doi: 10.22190/FUME230805029W.

ARTICLE IN PRESS